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Non-intrusive series arc fault detection for unknown scenarios based on differential current and synthetic arcing data

Wenpeng Luan, Hanju Cai, Bochao Zhao *, Yang Han

School of Electrical and Information Engineering, Tianjin University, 300072 Tianjin, China

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ABSTRACT

Series arc fault (SAF) detection is essential for the safety of residential power systems. However, in non-intrusive detection based on trunk currents, SAF signatures are typically masked by other loads operating in parallel. The difficulty of extracting robust fault features under such load interference and the data distribution shift between different domains make models trained in specific scenarios ineffective in unknown ones. To address this issue, this paper proposes a differential current extraction method based on dynamic baseline subtraction to provide signals with suppressed masking-load interference for SAF detection, improving the signal-to-noise ratio for learning intrinsic fault features and reducing the distribution gap across domains. Furthermore, an interpretable SAF data synthesis method based on dimension-wise class activation map is developed in synergy with differential currents to overcome the unavailability of real fault data in unknown scenarios for model adaptation. This method generates realistic SAF samples by fusing SAF signatures from the source domain with target-domain-specific background signals. Subsequently, a dual-branch neural network is employed as an effective classifier to validate the proposed methods, and it is further compressed through knowledge distillation via knowledge review to enable practical deployment on resource-constrained edge devices. Experimental results demonstrate that the proposed methods substantially enhance the cross-domain generalization, raising SAF detection accuracy in unknown scenarios from 66.2% to 96.4%. Moreover, the lightweight model achieves a detection time of 59.8 ms on the Raspberry Pi 4B, meeting the IEC 62606 standard with a negligible accuracy loss of 0.7% compared with the full-sized model. Notably, when integrated with existing methods, the proposed methods improve their average accuracy by 26.8% in unknown scenarios, demonstrating broad universality.

1. Introduction

The reliability of residential power systems is critical to personal and property safety. However, electrical fires pose a significant risk in these systems, and arc faults are widely recognized as a primary cause [1]. These faults typically arise from aging or damaged insulation, cable core breakage, and loose electrical connections, generating intense heat that can reach several thousand degrees Celsius and greatly increasing fire hazards [2]. Arc faults are generally categorized into series, parallel, and grounded faults. Parallel and grounded arc faults can be interrupted by traditional protection devices, such as circuit breakers and residual current devices, because the currents produced by these faults typically exceed the devices' operating thresholds [3]. However, series arc faults (SAFs) generate currents typically lower than normal operating levels, and their characteristics are highly dependent on the load connected in series. With the growing diversity of residential appliances, SAFs have become increasingly complex, posing a significant challenge to reliable detection [4].

In recent years, extensive research on SAF detection has been conducted, including physical-signal-based and electrical-signal-based methods. The former detect SAFs by monitoring light [5], heat [6], and electromagnetic radiation [7] generated by the faults. However, their applicability is primarily limited to fixed locations with potential risks, such as terminal connections in distribution rooms and switchgear. The latter detect SAFs by analyzing variations in voltage or current signals between arcing and normal samples. Although voltage-based methods have achieved promising performance in simple residential circuits [8,9], voltage fluctuations and load switching events may lead to false positives. Moreover, these techniques require further validation in more complex circuits. The current signals exhibit distinct SAF signatures, making current-based detection methods a major research focus. To quantify these signatures, various time-domain features have been

* Corresponding author.
E-mail address: bochaozhao@tju.edu.cn (B. Zhao).

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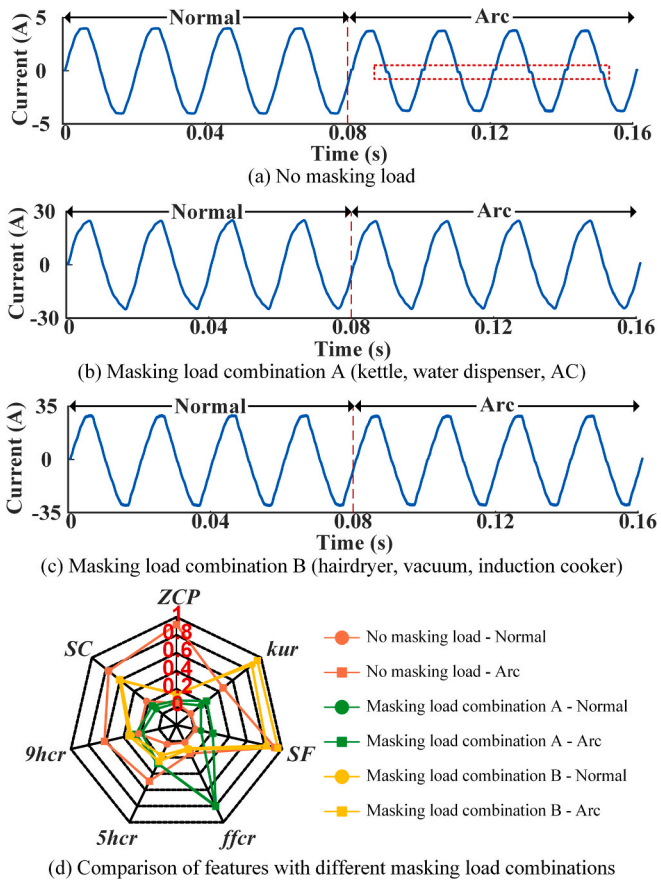


Fig. 1. Trunk currents and features under different load combinations.

developed, such as the zero-current period (*ZCP*), kurtosis (*kur*), shape factor (*SF*), error between adjacent current cycles, *L*2/*L*1 norm, and autocorrelation coefficient [10–14]. To comprehensively analyze SAF signatures, frequency domain and time–frequency domain transforms, such as the Fast Fourier Transform (FFT) [15], Chirp Z-Transform [16], Discrete Wavelet Transform [17], and Mode Decomposition [18], are utilized to extract a broader set of features, and the fusion of multi-domain features has proven effective in enhancing detection accuracy. To mitigate the negative effects of irrelevant features and reduce computational cost, feature engineering techniques are widely applied to select the most discriminative features [1,11,15,18–21].

Threshold-based criteria derived from the extracted expert features are initially employed for SAF detection due to their low computational burden and ease of implementation [12,13,16,17]. However, establishing robust thresholds is challenging in complex residential circuits owing to the diverse array of loads and their numerous simultaneous operating combinations. To this end, artificial intelligence (AI) techniques, such as decision trees [19,20], random forests (RFs) [11,14,18], k-nearest neighbors [1], fully connected neural networks [15], and convolutional neural networks (CNNs) [10], are applied to automatically establish detection boundaries. Given the heavy reliance on domain expertise and the limited representation capacity of expert features, end-to-end deep learning approaches have been increasingly adopted, which utilize neural networks to learn fault-related representations directly from raw current signals or their image and matrix transformations [22–28]. When the training and testing sets are independent and identically distributed (i.i.d.), which is a standard assumption in deep learning, such detection methods demonstrate remarkable performance.

However, in non-intrusive SAF detection based on trunk currents, when the SAF load operates in parallel with normal loads, which are referred to as masking loads, the observed currents deviate significantly from those in the absence of masking loads. Fig. 1 presents the trunk currents and the extracted features with and without masking loads when an SAF occurs in the branch of an electric caldron. As illustrated in Fig. 1(a), current shoulders remain observable in the absence of masking loads, but they become indistinguishable in the presence of different masking-load combinations, as shown in Fig. 1(b) and (c). For a more intuitive comparison, Fig. 1(d) presents a set of common features, including *ZCP*, *kur*, *SF*, spectral centroid (*SC*) [11], the fundamental frequency component rate (*ffcr*), and the fifth and ninth harmonic component rates (5*hcr,* 9*hcr*) [14]. These features differ substantially between normal and arcing samples when the masking loads are not involved, but they exhibit high similarity when the masking loads are connected in parallel. Therefore, features extracted from raw trunk current signals are tightly coupled with masking loads, making it difficult to maintain consistent fault representations across scenarios. Furthermore, the variability of load compositions often introduces a significant distribution shift between the data in the pre-existing training set and the unknown deployment scenarios, which violates the standard i.i.d. assumption between training and test data. Therefore, AI-based models trained in known scenarios are unable to guarantee stable performance when encountering samples with unknown load compositions [29,30]. Although exhaustive data collection could theoretically mitigate this generalization challenge, it is infeasible in practice.

To address these challenges, Jiang *et al.* [29] adopt a differential magnetic flux coupling method to isolate arcing signals from dominant trunk currents, thereby extracting regular SAF features and improving the model's generalization capability. However, since this method relies on capturing high-frequency components through MHz-level sampling, its effectiveness may be compromised by signal attenuation over long residential power lines. Transfer learning represents another promising strategy, which is predicated on the assumption that both SAF and normal data from the target domain are available for fine-tuning the pre-trained model [31]. Nevertheless, in residential arc fault detection, while normal data are readily accessible, collecting SAF data is impractical because the infrequent and stochastic nature of arc faults makes it both infeasible and hazardous. These limitations highlight a fundamental gap in existing residential SAF detection methods: although these methods can achieve satisfactory performance under controlled conditions, their deployment in real households remains constrained by the limited robustness of extracted features and the scarcity of SAF data in target deployment scenarios. Therefore, a more feasible solution should not only enhance discriminative SAF features under complex masking-load interference, but also enable detection model adaptation in the absence of target-domain SAF samples.

To this end, this paper presents a holistic framework for SAF detection in unknown scenarios, which uses differential currents to enhance feature generalization and synthetic SAF data to facilitate model adaptation without requiring target-domain fault samples. The main contributions of this paper are as follows:

(1) A differential current extraction method based on dynamic baseline subtraction is developed to isolate SAF-related signals from trunk currents while suppressing masking-load interference and preserving intrinsic fault signatures. This method enhances the domain invariance of the extracted features and improves model robustness across diverse load compositions.
(2) To address the scarcity of SAF data in unknown scenarios, an interpretable SAF differential current synthesis method based on dimension-wise class activation maps (dCAM) is developed to generate synthetic SAF samples with distributions close to real SAF data, thereby facilitating the detection model adaptation to unknown scenarios.
(3) A multi-level knowledge distillation method based on knowledge review is employed to compress the detection model for

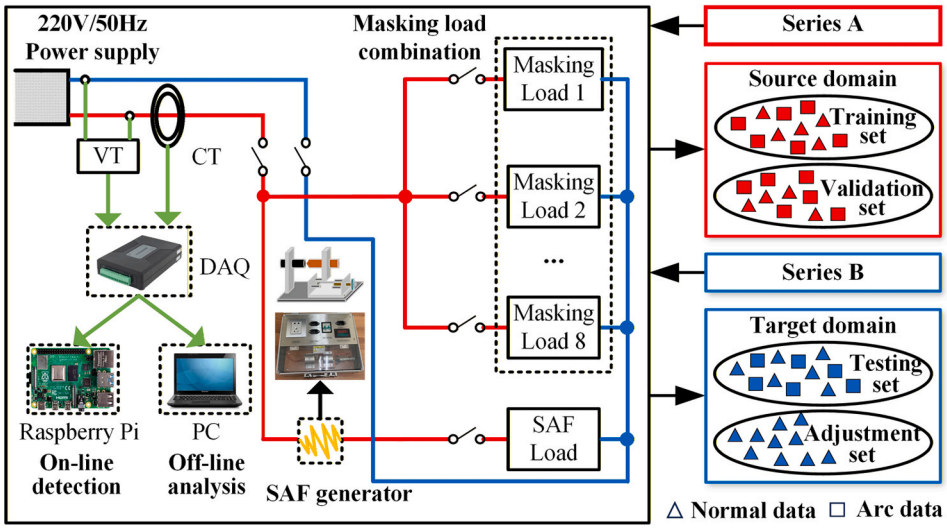


Fig. 2. SAF experimental platform diagram.

Table 1
Experimental loads in series A.

Type	Load – Brand	Type	Load – Brand	Type	Load – Brand	Type	Load – Brand
R	incandescent lamp – FSL	R	blender (heating) – Midea	IM	fan – Aucma	SPS	computer host – Lenovo
R	hairdryer (high heat) – Midea	R	water dispenser – Melng	IM	air conditioner – Haier	SPS	computer monitor – Dell
R	hairdryer (low heat) – Midea	SM	drill – Chnt	BM	hairdryer (high cool) – Midea	SPS	fluorescent lamp – Leesa
R	kettle – Changh	SM	blender (blending) – Midea	BM	hairdryer (low cool) – Midea	EC	induction cooker – Midea
R	electric caldron (high) – Changh	SM	vacuum cleaner – Slioper	SPS	dimmer lamps – Geeput	HWR	electric caldron (low) – Changh
R	halogen lamp – Osram	SM	juicer – Joyoung	SPS	power adapter – Flyco	MA	microwave oven – Smeg

deployment on resource-constrained edge devices. It transfers hierarchical knowledge from the teacher network to a lightweight student network, enabling real-time SAF detection with minimal accuracy loss.

(4) The proposed framework is validated with a dual-branch neural network classifier, achieving a detection accuracy of 96.4% in unknown scenarios. In addition, the framework demonstrates broad applicability by improving the average accuracy of existing models by 26.8% in unknown scenarios when integrated with them.

The rest of the paper is organized as follows. Section 2 details the configuration of the SAF experimental platform and the datasets. The proposed SAF detection method for unknown scenarios is introduced in Section 3. The experimental results and discussions are demonstrated in Section 4. Finally, Section 5 concludes the paper.

2. Experimental configuration and datasets

In this paper, the known scenario represents a residential environment where the load composition, defined as the specific set of loads available in the circuit, is predefined, and both normal and SAF data are reliably collected and labeled for supervised model training. In contrast, the unknown scenario represents another residential deployment environment where the load composition is unseen to the model trained in the known scenario, and only normal data are available for model adaptation. Given the inherent diversity of loads across different households, the unknown load composition consists of a set of loads that are distinct from those in the known scenario. To validate the generalization capability of the SAF detection model trained in the known scenario when deployed in an unknown scenario, the experimental platform is designed as shown in Fig. 2. This platform employs two independent load series, designated as Series A and Series B, to construct source-domain and target-domain datasets representing the known and unknown scenarios, respectively. The source-domain datasets are utilized for training and validation. Within the target-domain datasets, a portion of the normal data are allocated as an adjustment set for SAF synthesis, while the remaining data are reserved for testing.

As listed in Tables 1 and 2, both Series A and Series B consist of 19 appliances. Typically, most households share common appliances such as hairdryers and kettles, which may differ by brand, and also contain appliances unique to individual homes. To reflect this, the load composition in the unknown scenario differs from that in the known scenario in two aspects: (1) certain appliances in Series B correspond to those in Series A but are manufactured by different brands, and (2) the remaining appliances in Series B are absent from Series A. In both Series

Table 2
Experimental Loads in Series B.

Type	Load – Brand	Type	Load – Brand	Type	Load – Brand	Type	Load – Brand
R	hairdryer (high heat) – Flyco	R	halogen lamp – Sod	IM	fan – Melng	SPS	power adapter – Romoss
R	hairdryer (low heat) – Flyco	R	blender (heating) – Joyoung	IM	refrigerator – Midea	SPS	computer host – Dell
R	electric pan (high) – Changh	SM	drill – Delixi	IM	washing machine – Midea	SPS	television – Konka
R	electric pan (low) – Changh	SM	blender (blending) – Joyoung	BM	hairdryer (high cool) – Flyco	EC	induction cooker – Changh
R	rice cooker – Joyoung	SM	vacuum cleaner – Xiaomi	SPS	dimmer lamps – Geeput	HWR	hairdryer (low cool) – Flyco
R	kettle – Slioper	SM	juicer – Midea	SPS	humidifier – Aux	MA	microwave oven – Midea

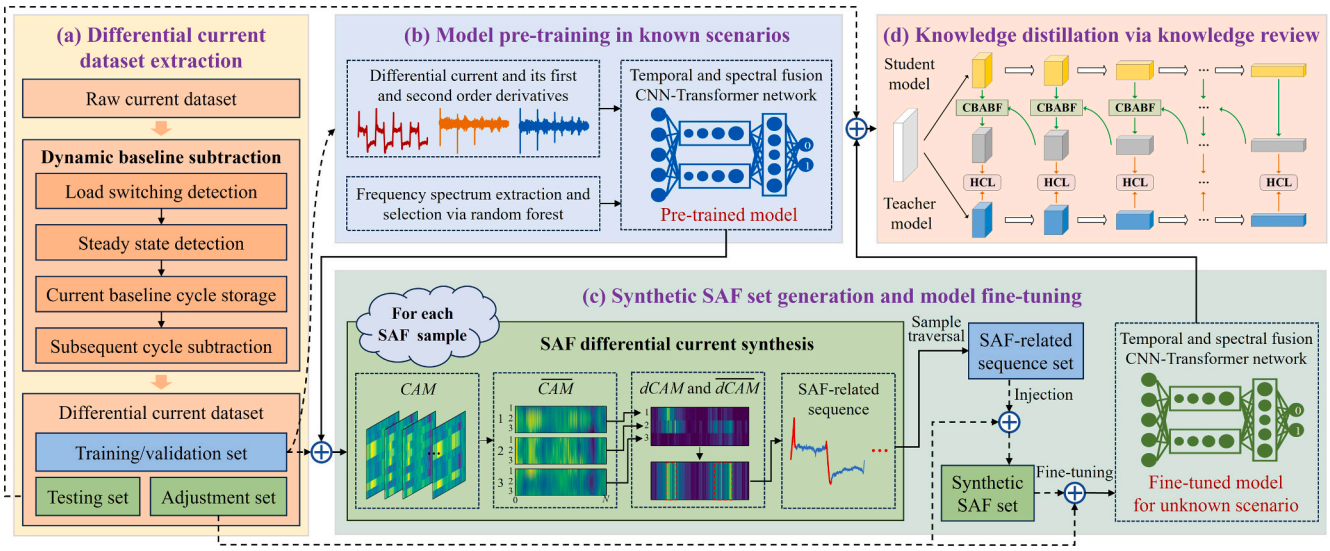


Fig. 3. Schematic of the proposed framework.

A and Series B, most appliances operate in a single state, except for the hairdryer with four states and the electric caldron, electric pan, and blender with two states each. As in [16], each operating state is treated as a separate load. Therefore, both Series A and Series B comprise 24 loads, covering eight load types: resistive (R), series motor (SM), induction motor (IM), brush motor (BM), switching power supply (SPS), eddy current (EC), half-wave rectifier (HWR), and magnetron (MA). These eight types cover most typical household loads, and loads within the same type share similar operating principles.

For both Series A and Series B, the experiments are designed to collect data from 24 single-branch circuits featuring an individual load and 140 multi-branch circuits. In each multi-branch circuit, one load is randomly selected as the SAF load, while up to eight additional loads serve as masking loads. The loads are switched on sequentially, and the SAF is generated in the branch of the SAF load by an arc fault generator specified in UL 1699 [32] after all switched-on loads reach steady-state operation for a designated period. The trunk current and terminal voltage are measured using an HCS-ES5 Hall current transformer (CT) and an HVSAS Hall voltage transformer (VT), respectively. These measurements are initially collected at a 50 kHz sampling rate using a USB-3200 data acquisition (DAQ) card and are subsequently downsampled to 12.5 kHz. To ensure that the data are sufficiently comprehensive for analysis, the collection process covers the entire operational sequence, including load switching events, subsequent transient currents, and steady-state currents.

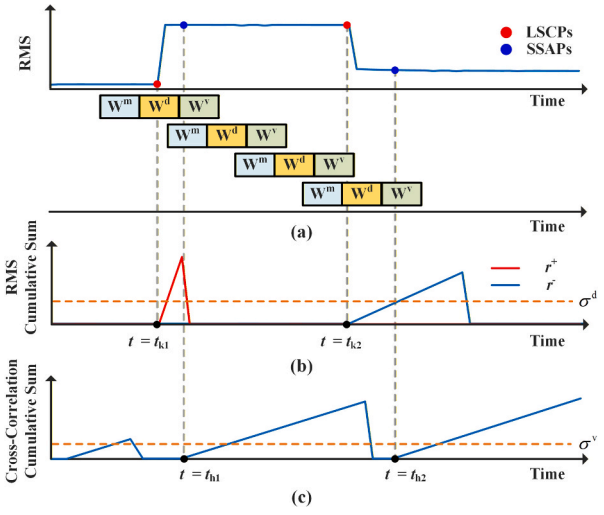


Fig. 4. Schematic diagram of LSCP and SSAP detectors.

3. Proposed SAF detection method

The proposed framework, illustrated in Fig. 3, comprises four stages. First, dynamic baseline subtraction is applied to the raw trunk current to extract the differential current datasets. In this procedure, load switching events and the subsequent onset of steady-state operation are detected to dynamically update the baseline cycle, which is then subtracted from the incoming raw current waveforms. Second, a temporal-spectral fusion network integrating CNN and Transformer architectures is pre-trained utilizing the differential current data from the known scenario. In the third stage, to adapt the model to the unknown scenario where SAF data are unavailable, SAF differential current synthesis is performed. Specifically, SAF-related sequences are extracted from the known scenario based on dCAM and injected into the adjustment set to create a synthetic SAF set for the unknown scenario. The pre-trained network is subsequently fine-tuned using both the adjustment set and the synthetic SAF set. Finally, to facilitate practical deployment, the fine-tuned network is compressed into a lightweight network through knowledge distillation via knowledge review, and its performance is then evaluated on the Raspberry Pi 4B.

3.1. Differential current waveform extraction

The raw trunk current I^{raw} consists of the intrinsic arc fault signature F^{arc}, which the detection model needs to capture, and the background component S^{mask}. Although arc-related features vary across SAF loads, loads within the same type generally exhibit common operating mechanisms and share similar intrinsic SAF signatures. For example, resistive loads tend to present flat shoulders, while SPS loads typically produce sharp pulses. As a result, the intrinsic SAF signatures in known and unknown scenarios, denoted as F_A^{arc} and F_B^{arc}, remain relatively similar. In contrast, the background component S^{mask}, which is determined by parallel masking loads, is highly variable across scenarios, and often has a much larger amplitude than the relatively weak F^{arc}. Therefore, in non-intrusive SAF detection based on trunk currents, the intrinsic arc fault signatures often exhibit an extremely low signal-to-noise ratio (SNR) due to interference from the background component. The feature representation capability of deep learning models typically degrades when processing such highly noisy signals, making it difficult to extract consistent

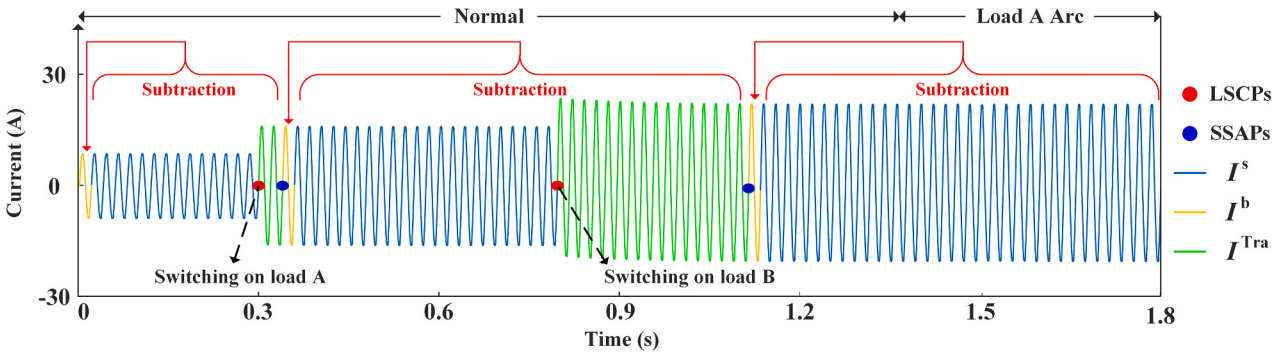


Fig. 5. Schematic diagram of differential current waveform extraction.

SAF features and leading to poor domain invariance in the learned representations [33]. Moreover, the statistical distribution of the trunk currents is heavily dictated by S^{mask}, and the data distribution in the known scenario, denoted as $\mathscr{D}\left(F_A^{\text{arc}}, S_A^{\text{mask}}\right)$, exhibits a distributional shift relative to that in the unknown scenario $\mathscr{D}\left(F_B^{\text{arc}}, S_B^{\text{mask}}\right)$. The difficulty of capturing the intrinsic arc fault feature, together with the significant data distribution shift between the source and target domains, makes detection models trained in known scenarios ineffective when deployed in unknown scenarios.

To mitigate the interference of masking loads in non-intrusive SAF detection, a differential current extraction method based on dynamic baseline subtraction is proposed. The method first identifies load switching change points (LSCPs) and the subsequent initial steady-state anchoring points (SSAPs) using cost-effective LSCP and SSAP detectors, respectively. Given the trunk current, three cascaded windows, namely an L-length window W^{m}, a K-length window W^{d}, and an H-length window W^{v}, are employed to analyze its cycle-by-cycle root mean square (RMS) values. Here, W^{d} and W^{v} are used to identify the LSCPs and SSAPs, respectively, as shown in Fig. 4(a). To identify LSCPs caused by load switching events, the RMS cumulative sum statistics are defined as r_+^k and r_-^k:

$$r_k^+ = \begin{cases} 0, & \text{if } k = 0 \text{ or } \mu^{\mathrm{d}} - (\mu^{\mathrm{m}} + \theta) < 0 \\ r_{k-1}^+ + \mu^{\mathrm{d}} - (\mu^{\mathrm{m}} + \theta), & \text{otherwise} \end{cases} \tag{1}$$

$$r_k^- = \begin{cases} 0, & \text{if } k = 0 \text{ or } \mu^{\mathrm{d}} - (\mu^{\mathrm{m}} - \theta) > 0 \\ r_{k-1}^- - \mu^{\mathrm{d}} + (\mu^{\mathrm{m}} - \theta), & \text{otherwise} \end{cases} \tag{2}$$

where μ^{m} and μ^{d} represent the RMS means in W^{m} and W^{d}, respectively, each $k \in [1, K]$ denotes the index, and θ is the RMS stability margin. The LSCP detection rule is defined as $r_k^+ > \sigma^{\mathrm{d}}$ or $r_k^- > \sigma^{\mathrm{d}}$, where σ^{d} is a preset threshold. The timestamps of the LSCPs are backdated to the initial zero-accumulation onset, as exemplified by t_{k1} and t_{k2} in Fig. 4(b).

Following the LSCPs, the current typically enters a transient phase before stabilizing into a steady state, exhibiting a transient current I^{Tra}. To further identify the following SSAPs, the cross-correlation cumulative sum statistic g_h is defined as:

$$g_h = \begin{cases} 0, & \text{if } h = 0 \text{ or } \left|C^{\mathrm{m}} - C^{\mathrm{d}}\right| > \varepsilon \\ g_{h-1} + C^{\mathrm{m}} + C^{\mathrm{d}}, & \text{otherwise} \end{cases} \tag{3}$$

where C^{m} denotes the cross-correlation coefficient between the RMS values in W^{m} and W^{v}, and C^{d} denotes that between the RMS values in W^{d} and W^{v}; $h \in [1, H]$ is the index, and ε represents the deviation threshold.

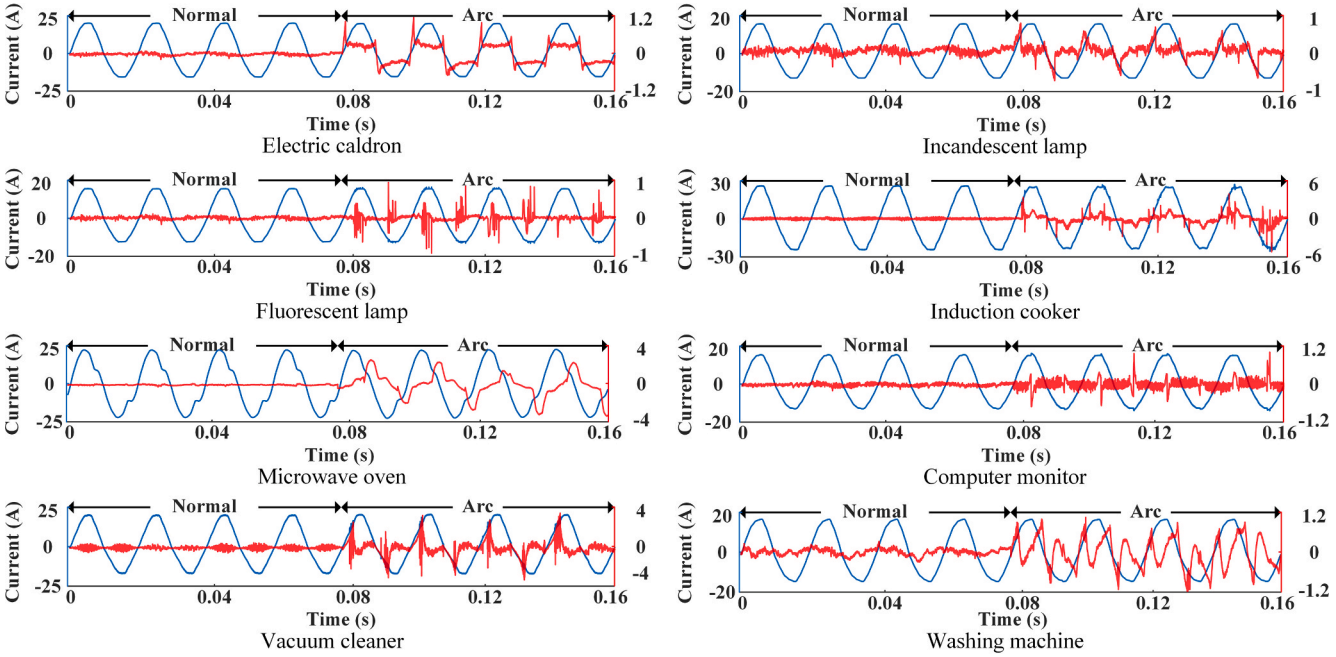


Fig. 6. The raw and differential current waveforms of appliances with masking loads.

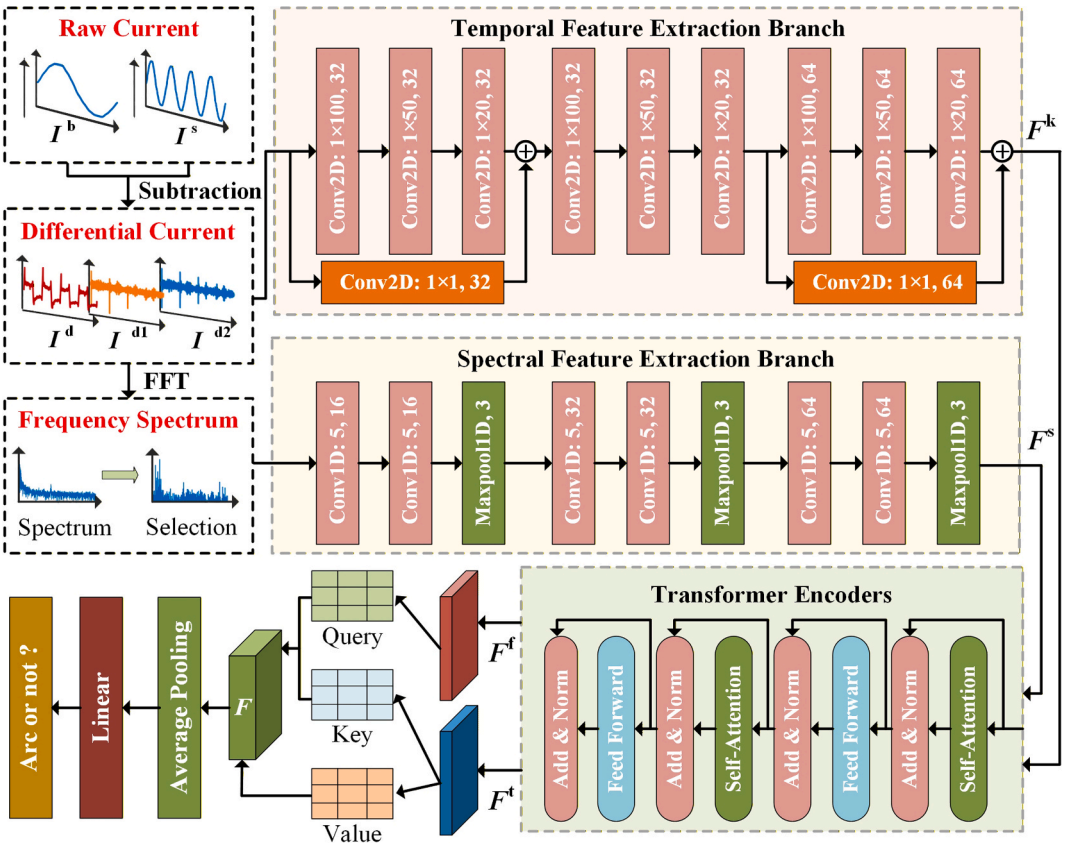


Fig. 7. Architecture of ArcFusionNet.

When g_h exceeds the threshold σ^v, the SSAP is detected with its time-stamp backdated to the initial zero-accumulation onset, as exemplified by t_{h1} and t_{h2} in Fig. 4(c).

To better illustrate the differential current extraction method, Fig. 5 presents an example of an SAF event when multiple loads operate simultaneously, where the LSCPs and SSAPs are indicated by red and blue points, respectively. The n-length current cycle corresponding to the SSAP serves as the baseline $I^{b} = \left[i_1^{b}, i_2^{b}, \ldots, i_n^{b}\right]$, while the subsequent cycles function as the subtracted waveform $I^{s} = \left[i_1^{s}, i_2^{s}, \ldots, i_n^{s}\right]$. Both I^{b} and I^{s} are phase-aligned according to the voltage waveform, and the differential current cycle I^{d} is then calculated as follows:

$$I^{d} = I^{b} - I^{s} = \left[i_1^{b} - i_1^{s}, i_2^{b} - i_2^{s}, \ldots, i_n^{b} - i_n^{s}\right] \tag{4}$$

It should be noted that the differential baseline for I^{Tra} is the cycle preceding load switching. In this case, $I^{d} = I^{q} + \beta$, where I^{q} represents the current of the single load being switched, and β denotes the residual noise caused by scenario-specific voltage and current fluctuations, measurement noise, and other perturbations. Given that SAF detection methods for single load have been well validated in the existing literature, SAF detection during the transient phase is regarded as a separate problem beyond the primary scope of this work. Accordingly, the monitoring objective is defined as the subsequent steady-state current. Therefore, it is reasonably assumed that the transient phase concludes without faults, thereby designating the initial steady-state cycle I^{b} as a normal baseline. For subsequent SAF detection, the differential current is defined as:

$$I^{d} = \begin{cases} \beta, & \text{if } I^{s} \text{ is normal} \\ I^{c_normal} - I^{c_arc} + \beta, & \text{if } I^{s} \text{ is arc} \end{cases} \tag{5}$$

where I^{c_normal} and I^{c_arc} denote the raw currents of the load connected in series with the SAF under normal and arcing conditions, respectively.

The differential current extraction method functions as a linear transformation to filter out the dominant background component S^{mask}, resulting in the transformed data distributions $\mathscr{D}\left(F_A^{arc_d}, \beta_A^{d}\right)$ and $\mathscr{D}\left(F_B^{arc_d}, \beta_B^{d}\right)$ in the known and unknown scenarios, respectively. Here, F^{arc_d} represents the arc signatures of the differential current and β^{d} denotes the subtle residual noise. This transformation is theoretically important as it improves the SNR of the arc signatures by removing heavy masking-load interference, thereby enabling the model to more effectively capture the intrinsic features associated with arc faults. Furthermore, the distributional shift is reduced because the domain gap is no longer dominated by overwhelming masking loads. Instead, it stems primarily from minor variations between the residual noise components β_A^{d} and β_B^{d}, as well as from slight differences between the intrinsic arc signatures $F_A^{arc_d}$ and $F_B^{arc_d}$.

To visually demonstrate the effectiveness of this extraction method, Fig. 6 presents the raw trunk currents and differential currents for eight typical loads under the same masking load as those in Fig. 5. In each subplot, the blue line corresponds to the raw trunk current on the left y-axis, and the red line represents the differential current on the right y-axis. As indicated by the blue traces, the high-amplitude current from the masking loads dominates the trunk current and makes the subtle distortions caused by SAFs nearly imperceptible, highlighting the difficulty of directly detecting SAFs from the raw trunk current in complex scenarios. In contrast, the differential current effectively decouples the SAF signatures from the masking loads. During normal operation, the differential current remains a low-amplitude and noise-like waveform that fluctuates around zero. However, when the SAF occurs, it reveals distinct arc fault signatures, including high-frequency transients such as sharp spikes and pulses, and low-frequency patterns such as saddle-shaped waveforms, which correspond to the flat shoulder in the raw current waveform.

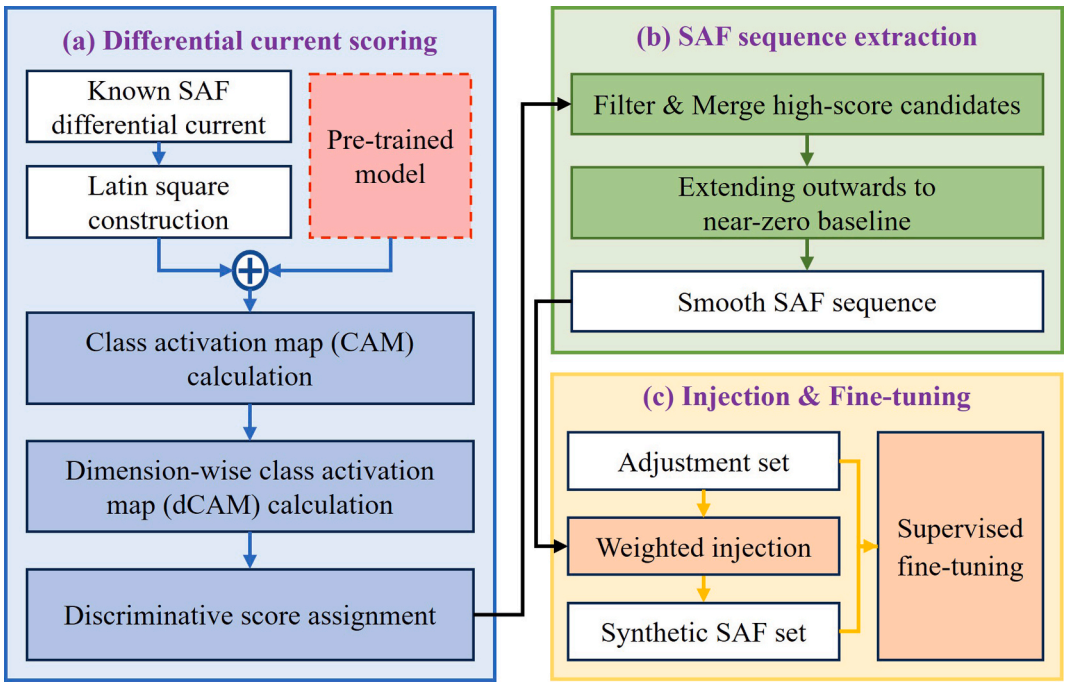


Fig. 8. Flowchart of the SAF differential current synthesis and model fine-tuning strategy.

3.2. ArcFusionNet: temporal-spectral fusion network based on differential current

An effective classifier is required to process the differential current and capture both the high-frequency and low-frequency SAF signatures. Since these signatures are manifested differently in the temporal and spectral domains, ArcFusionNet is designed to jointly exploit temporal and spectral features for SAF detection, as shown in Fig. 7. The architecture adopts a hybrid CNN-Transformer framework, in which CNN-based temporal and spectral branches are first used to extract local features, followed by Transformer encoders for global dependency modeling. Specifically, the temporal branch focuses on local waveform patterns, while the spectral branch captures harmonic characteristics. The resulting representations are then fed into Transformer encoders to model their global relationships. This hierarchical design enables progressive modeling from local feature extraction to global dependency learning.

The temporal input is defined as a one-cycle segment of the differential current I^{d}. This length is a widely adopted configuration in the existing literature, and enables the model to capture full-cycle structural information, such as inter-half-cycle asymmetry, which is important for reliable SAF detection. It also complies with the half-cycle counting logic specified in IEC 62606 [34] while maintaining a relatively low computational cost. In addition to I^{d}, its first-order and second-order derivatives, I^{d_1} and I^{d_2}, are also incorporated as inputs to the temporal feature extraction branch. The introduced derivative inputs enable the model to capture information related to the rate of change and local curvature of the differential current, thereby enriching the temporal representation. The temporal feature extraction branch consists of two residual blocks with a convolutional block inserted between them. Each block contains three convolutional layers with kernel sizes of 1×100, 1×50, and 1×20, respectively. The first residual block and the convolutional block output 32 channels, whereas the second residual block outputs 64 channels.

The input to the spectral branch is obtained by applying the FFT to the one-cycle differential current. While the resulting spectrum reflects the harmonic composition of the signal, it is high-dimensional and may contain redundant or irrelevant information. Directly using all harmonic components as inputs increases computational complexity and may introduce noise that degrades model performance and generalization. Therefore, a feature selection step is performed to identify the most discriminative harmonic components associated with SAFs. Specifically, the RF algorithm is used to estimate the importance of each harmonic component, and the resulting importance scores serve as the criterion for feature selection. The selected harmonic components are then fed into the spectral branch, which consists of three blocks. Each block contains two Conv1D layers with a kernel size of 5, followed by a Maxpool1D layer with a pool size of 3. The Conv1D layers output 16, 32, and 64 channels in the first, second, and third blocks, respectively.

Each convolutional layer in the branches employs zero-padding to preserve the feature dimensions and is followed by ReLU activation and Batch Normalization. The resulting temporal representation F^{k} and spectral representation F^{s} are then processed separately by Transformer encoders to model their global dependencies, yielding the encoded representations F^{t} and F^{f}, respectively. A cross-attention mechanism is subsequently used to fuse these two representations. During this fusion process, F^{f} serves as the query while F^{t} serves as the key and value. The attention weights capture complementary information from the temporal and spectral domains, and the fused representation F is computed as follows:

$$F = \text{Attention}\left(F^{f}, F^{t}, F^{t}\right) = \text{softmax}\left(\frac{F^{f}\left(F^{t}\right)^{\mathrm{T}}}{\sqrt{d^{t}}}\right)F^{t} \tag{6}$$

where d^{t} represents the dimension of the temporal representation. After the cross-attention operation, an adaptive average pooling layer is applied to aggregate the fused representation, followed by a linear layer for binary SAF classification.

3.3. SAF differential current synthesis for ArcFusionNet adaptation

The proposed ArcFusionNet is first pre-trained in the known scenario using labeled normal and SAF data under the supervision of a cross-entropy loss function. To adapt the model to the unknown scenario without target-domain SAF data, an interpretable data synthesis method is proposed to generate synthetic SAF samples for model fine-tuning.

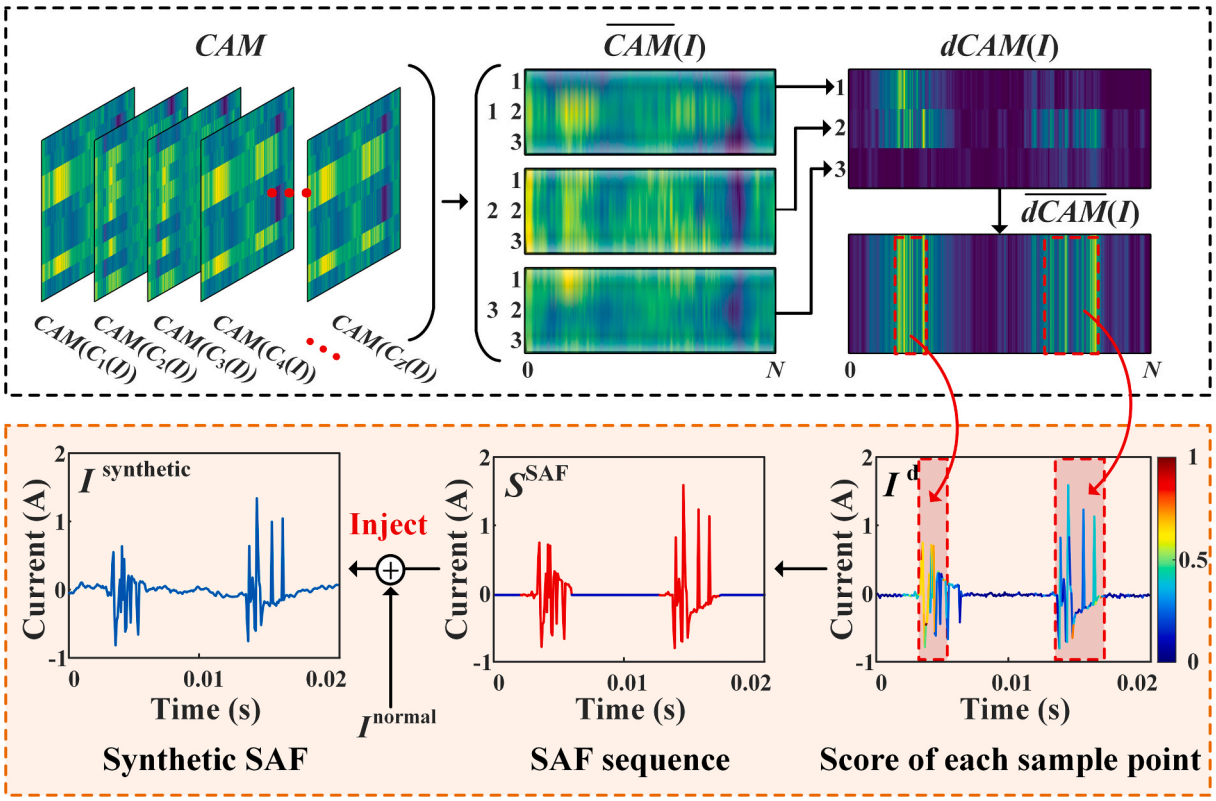


Fig. 9. The procedure of dCAM for SAF differential current synthesis.

The overall flowchart is illustrated in Fig. 8. First, the knowledge embedded in the pre-trained model is leveraged via dCAM [36] to assign an importance score to each sampling point within each differential current sample from the source domain. Subsequently, high-scoring candidates undergo post-processing operations, including filtering, merging, and extending, to extract the sequences related to the arc signatures. This process is iteratively performed on each source-domain SAF differential current sample to obtain a comprehensive set of smooth SAF-related sequences representing the arc signatures $F_A^{\text{arc_d}}$. Then, these sequences are injected into the differential normal data from the target domain, which provides an effective sampling of the distribution of target residual noise components β_B^{d}. Therefore, this injection process generates a synthetic SAF set with a distribution approximated as $\mathscr{D}\left(F_A^{\text{arc_d}}, \beta_B^{\text{d}}\right)$. The distribution shift relative to the real target distribution $\mathscr{D}\left(F_B^{\text{arc_d}}, \beta_B^{\text{d}}\right)$ is thus mainly attributed to the minor difference between the arc signatures $F_A^{\text{arc_d}}$ and $F_B^{\text{arc_d}}$.

The entire procedure for SAF differential current synthesis begins by scoring individual sample points of I^{d} based on dCAM, as illustrated in Fig. 9. Given the input data $\boldsymbol{I} = \left[I^{\text{d}}, I^{\text{d}_1}, I^{\text{d}_2}\right]$, a new structure $C_z(\boldsymbol{I}) \in \mathbb{R}^{3\times3\times n}$ is constructed, whose first two dimensions form a third-order Latin square using elements from $\boldsymbol{I}$. Let Z denote the total number of valid Latin squares. Then, the class activation map is expressed as a function of $C_z(\boldsymbol{I})$:

$$CAM(C_z(\boldsymbol{I})) = \sum_{j=1}^{Q} W_j^c \cdot F_j^k(C_z(\boldsymbol{I})) \in \mathbb{R}^{3\times3\times n} \tag{7}$$

where $z \in [1, Z]$ denotes the index, $F_j^k(C_z(\boldsymbol{I}))$ denotes the representation of the j-th channel from the last residual block in the temporal branch, and Q denotes the total number of channels in F^k, which is equal to the number of neurons in the linear layer. The variable W_j^c represents the j-th weight of the linear layer corresponding to the binary category c, which indicates either an SAF or a normal sample. The $dCAM$ is then derived as:

$$dCAM(\boldsymbol{I}) = \mu(\overline{CAM}(\boldsymbol{I}), 2) \times \sigma^2(\overline{CAM}(\boldsymbol{I}), 2) \in \mathbb{R}^{3\times n} \tag{8}$$

where $\overline{CAM}(\boldsymbol{I}) \in \mathbb{R}^{3\times3\times n}$ denotes the mean of $CAM(C_z(\boldsymbol{I}))$ over $z = 1, \ldots, Z$, while $\mu(\overline{CAM}(\boldsymbol{I}), 2)$ and $\sigma^2(\overline{CAM}(\boldsymbol{I}), 2)$ denote the mean and variance of $\overline{CAM}(\boldsymbol{I})$ computed along the second dimension, respectively.

The mean of $dCAM(\boldsymbol{I})$ computed along the first dimension, denoted as $\overline{dCAM}(\boldsymbol{I}) \in \mathbb{R}^n$, assigns a discriminative score to each sample point, which guides the selection of SAF-related sample point candidates. These initial candidates may be spatially dispersed and are subsequently refined through filtering and merging operations to form one or more contiguous core regions, which represent the most salient parts of the SAF signatures. To prevent spurious discontinuities in the synthetic waveforms, a smooth expansion method is applied to obtain the SAF-related sequences S^{SAF} by extending the refined core regions both forward and backward along the original differential current waveform. The expansion continues until the amplitude of the waveform naturally decays and merges into the near-zero baseline, thereby ensuring seamless transitions at both edges.

The extracted S^{SAF} is subsequently injected into the normal differential current cycle I^{normal} obtained from the adjustment set, effectively integrating the SAF signatures with β_B^{d} in the target domain. To regulate the influence of the injected SAF-related sequence, a constant weighting factor $w \in [0, 1]$ and a small random noise component δ are introduced, and the synthetic SAF differential current cycle $I^{\text{synthetic}}$ is formulated as:

$$I^{\text{synthetic}} = wS^{\text{SAF}} + (1 - w)I^{\text{normal}} + \delta \tag{9}$$

Through iterative execution of this synthesis procedure, a synthetic SAF set is generated. This set, together with the adjustment set in the target domain, is used to fine-tune the pre-trained ArcFusionNet. In this way, the model can be adapted to the target domain without access to its real SAF data.

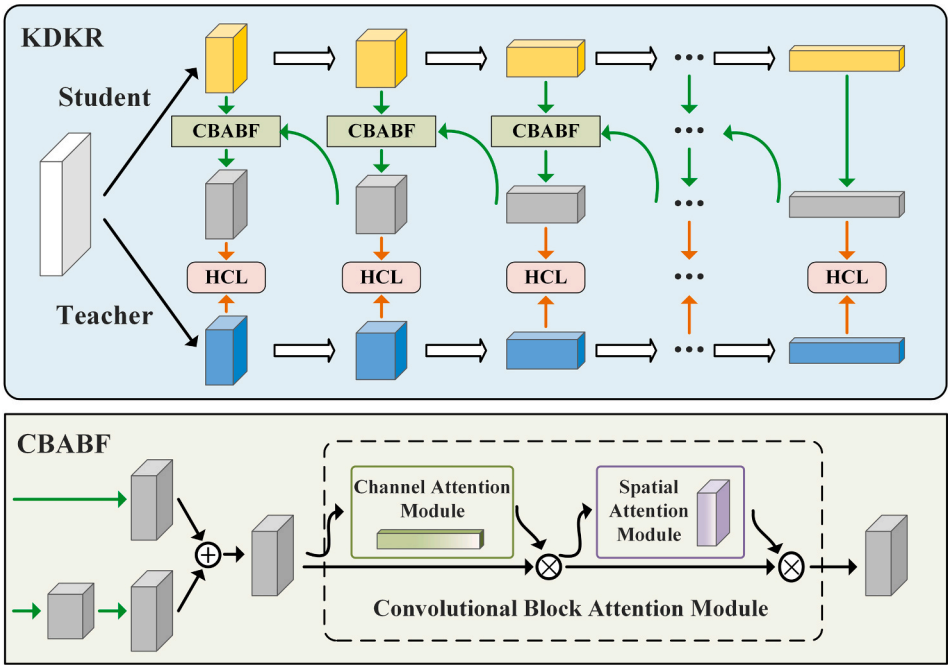


Fig. 10. Knowledge distillation for ArcFusionNet with structural details of CBABF.

3.4. *Knowledge distillation via knowledge review for ArcFusionNet*

Although the dual-branch architecture of the proposed ArcFusionNet is effective for temporal-spectral feature learning, its structural complexity inherently results in substantial computational and memory demands, which pose a challenge for deployment on resource-constrained edge devices. To balance model complexity and practical deployability, knowledge distillation via knowledge review (KDKR) is employed to transfer knowledge from a large teacher model to a lightweight student network, as illustrated in Fig. 10. KDKR guides the student network to learn not only from the output logits of the teacher model but also from the hierarchical representations of its intermediate layers.

To facilitate hierarchical feature aggregation in KDKR, a convolutional block attention-based fusion (CBABF) module is proposed. This module merges the input representations through element-wise addition, and the resulting combined representation is subsequently fed into a convolutional block attention module for further enhancement. Within the attention block, a channel attention module is applied to adaptively reweight the feature channels, while a spatial attention module is used to highlight the most informative locations. The final output of CBABF is a refined feature representation, in which key information is selectively emphasized across both the channel and spatial dimensions. Then, the hierarchical context loss (HCL) [36] is employed to quantify the representation discrepancy between the teacher and the student at each level, and the distillation loss $\mathscr{L}^{\text{dis}}$ is formulated as:

$$\mathscr{L}^{\text{dis}} = \mathscr{H}\left(F_M^{\text{stu}}, F_M^{\text{tchr}}\right) + \sum_{m=M-1}^{1} \mathscr{H}\left(\mathscr{U}\left(F_m^{\text{stu}}, F_{m+1,M}^{\text{stu}}\right), F_m^{\text{tchr}}\right) \tag{10}$$

where M denotes the total number of layers in the teacher and student networks, and $m \in [1, M]$ is the layer index. The terms F_m^{tchr} and F_m^{stu} represent the representations from the m-th layers of the teacher and student networks, respectively. $F_{m+1,M}^{\text{stu}}$ denotes the recursively fused feature representation from F_{m+1}^{stu} to F_M^{stu}. The operator $\mathscr{U}(\bullet)$ denotes the CBABF module designed for multi-level representation fusion, and $\mathscr{H}(\bullet)$ corresponds to the HCL module used to calculate the representation discrepancy.

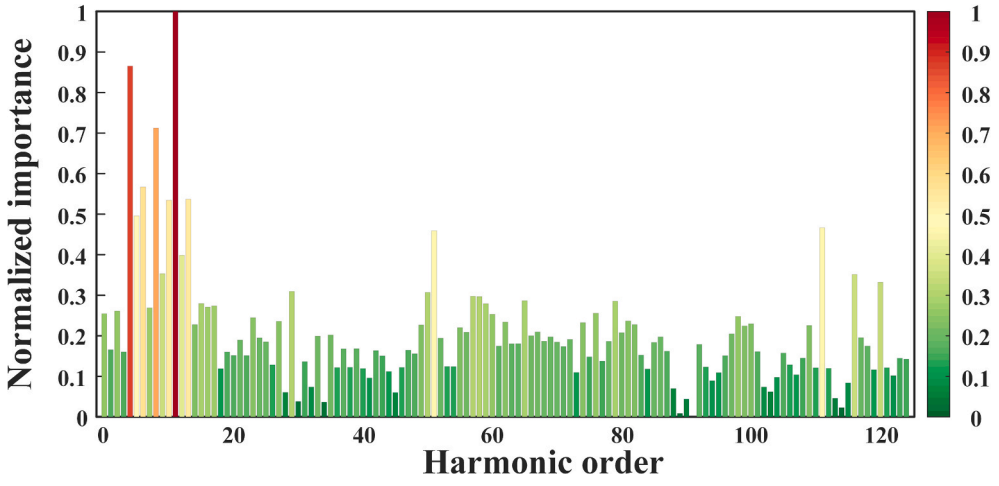


Fig. 11. The normalized importance scores of harmonic components.

Table 3
Specifications of differential current datasets.

Domain	Dataset	Normal sample number	SAF sample number
Source (known scenario)	Training set	128,220	214,938
	Validation set	32,539	53,251
Target (unknown scenario)	Testing set	17,966	31,215
	Adjustment set	71,864	/
/	Synthetic SAF set	/	143,728

The total loss $\mathscr{L}^{\text{total}}$ for training the student network is defined as a weighted sum of the cross-entropy loss $\mathscr{L}^{\text{CE}}$ and the distillation loss $\mathscr{L}^{\text{dis}}$:

$$\mathscr{L}^{\text{total}} = \alpha\mathscr{L}^{\text{CE}} + (1-\alpha)\mathscr{L}^{\text{dis}} \tag{11}$$

where α is a hyperparameter that balances the contributions of the two loss functions. During inference, only the lightweight student network is employed, resulting in no additional computational cost during testing.

4. Experimental analysis and verification

4.1. Experimental setup

In this paper, the window lengths for W^m, W^d, and W^v are uniformly set to 10, and the thresholds σ^d and σ^v are set to 10 and 55. The hyperparameters θ, ε, α, and w are set to 0.01, 3e-5, 0.5, and 0.6, respectively. The experiments are implemented in Python 3.10.6 and executed on a server equipped with a 12th-generation Intel Core i7-12700 CPU, 32 GB of RAM, and a single GeForce RTX 4060 GPU. The proposed network is trained for 70 epochs using the AdamW optimizer with a batch size of 64. An adaptive learning rate strategy is employed, with the learning rate initialized at 1e-4 and decayed to a minimum of 1e-6. Furthermore, the synthetic SAF set is generated to be twice the size of the adjustment set. The embedded implementation is deployed on the Raspberry Pi 4B equipped with a 1.5 GHz 64-bit quad-core Arm Cortex-A72 CPU and 8 GB of RAM.

The normalized importance scores of the harmonic components, as evaluated by the RF algorithm on the training set, are shown in Fig. 11. The analysis indicates that low-frequency harmonic components within the differential currents exhibit higher importance, while only a small fraction of high-frequency components achieve scores close to 0.5. This concentration of informative content in the low-frequency spectrum offers two practical advantages. First, it holds the potential to reduce the demand for high-frequency data sampling. Second, compared with high-frequency signals, low-frequency signals are less susceptible to attenuation over power lines, which makes their discriminative fault signatures more reliably detectable in non-intrusive SAF detection. Given that most harmonic components exhibit low importance scores, the top 32 harmonic components with the highest scores are selected as inputs to the spectral branch of ArcFusionNet.

The differential current datasets are constructed by applying the dynamic baseline subtraction method to the raw current data in both the source and target domains. The resulting differential current datasets include differential current samples derived from raw current in cases of missed SSAP detection. Such cases typically occur when a low-power load switching event does not produce an amplitude variation large enough to trigger the LSCP detector, leaving the subtraction baseline unupdated. In these situations, the extracted differential current naturally corresponds to the raw current of the switched load and remains independent of the masking loads. The detailed specifications of these datasets are summarized in Table 3. To evaluate detection performance, six widely used metrics are employed, namely accuracy (*ACC*), false negative rate (*FNR*), false positive rate (*FPR*), precision, recall, and *F*1-score, with their definitions following [1,31].

4.2. Cross-domain evaluation

To comprehensively evaluate the proposed method, its detection performance is assessed in both known (validation set in the source domain) and unknown (testing set in the target domain) scenarios. Furthermore, the proposed method is benchmarked against nine existing methods. Specifically, two of these benchmarks employ machine learning with expert features derived from raw currents [11,18], and the other seven adopt end-to-end neural networks based on raw currents or their various transformations [22–28]. The evaluation results are presented in Table 4.

In the known scenario, all benchmarks exhibit highly competitive performance, which is primarily attributed to the identical data distributions of the training and validation sets. Despite this strong overall baseline, the proposed ArcFusionNet demonstrates superior performance across all evaluation metrics, thereby validating its effectiveness as a classifier.

Regarding performance in the unknown scenario, the proposed method maintains a robust accuracy of 96.4%, validating its strong generalization capability. Error analysis indicates that false negatives mainly arise from weak SAFs, whose differential currents closely resemble those of the normal samples. The false positives primarily occur in multi-branch circuits, where substantial background noise from masking loads distorts normal currents to exhibit arc-like signatures. However, all benchmarks suffer a substantial performance degradation, with accuracy decreasing by 29.4% to 36.1%. Although their *FNR* increases only slightly, their *FPR* rises sharply. This discrepancy indicates that the models do not lose their capacity to detect arcs. Instead, they severely misclassify target-domain normal samples as source-domain SAFs due to the distributional shift driven by variations in load compositions. An intuitive countermeasure for mitigating the high *FPR* is to fine-tune the model using a combination of source-domain SAFs and target-domain normal data. However, as shown by the ablation study in

Table 4
Evaluation results in different scenarios.

Ref.	Known scenario (validation set in source domain)						Unknown scenarios (testing set in target domain)					
	ACC (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)	ACC (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)
[11]	97.6	2.1	2.9	97.5	97.5	97.5	65.1	10.3	73.3	64.0	58.2	61.0
[18]	97.9	1.6	2.8	97.9	97.8	97.9	62.5	4.8	88.3	61.9	53.5	57.4
[22]	97.3	2.6	2.9	97.1	97.3	97.2	67.2	4.0	77.6	72.0	59.2	65.0
[23]	96.7	3.3	3.4	96.3	96.7	96.5	63.8	4.4	85.6	65.7	55.0	59.9
[24]	96.8	3.4	2.9	96.5	96.8	96.7	65.9	9.6	72.3	65.6	59.1	62.1
[25]	98.4	1.7	1.5	98.2	98.4	98.3	65.8	2.4	83.9	73.0	56.9	64.0
[26]	96.4	3.9	3.2	96.1	96.5	96.2	67.0	4.5	77.5	71.1	59.1	64.5
[27]	97.1	2.8	2.9	96.9	97.1	97.0	65.0	4.4	82.5	68.0	56.5	61.7
[28]	98.0	2.2	1.7	97.8	98.1	97.9	61.9	**0.9**	96.0	68.3	51.6	58.8
Proposed	**98.9**	**1.3**	**0.8**	**98.8**	**99.0**	**98.9**	**96.4**	3.5	**3.8**	**95.9**	**96.4**	**96.1**

Table 5
Evaluation results of ablation studies.

Ablation Study	ACC (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)
Case (a)	91.3 (↓ 5.1)	8.3 (↑ 4.8)	9.4 (↑ 5.6)	90.3 (↓ 5.6)	91.1 (↓ 5.3)	90.7 (↓ 5.4)
Case (b)	95.6 (↓ 0.8)	4.6 (↑ 1.1)	4.1 (↑ 0.3)	94.9 (↓ 1.0)	95.6 (↓ 0.8)	95.2 (↓ 0.9)
Case (c)	84.5 (↓ 11.9)	25.2 (↑ 21.7)	0.3 (↓ 3.5)	85.7 (↓ 10.2)	87.2 (↓ 9.2)	86.4 (↓ 9.7)
Case (d)	66.2 (↓ 30.2)	2.0(↓ 1.5)	83.2 (↑ 79.4)	74.6 (↓ 21.3)	57.4 (↓ 39.0)	64.9 (↓ 31.2)
Case (e)	90.8 (↓ 5.6)	6.4 (↑ 2.9)	14.1 (↑ 10.3)	90.3 (↓ 5.6)	89.7 (↓ 6.7)	90.0 (↓ 6.1)
Case (f)	83.7 (↓ 12.8)	24.8 (↑ 21.3)	1.6 (↓ 2.2)	84.2 (↓ 11.8)	86.8 (↓ 9.6)	85.4 (↓ 10.7)

Section 4.3, this strategy remains insufficient.

4.3. Ablation studies

To assess the specific contribution of each proposed component, the ablation study defines six evaluation cases for analysis. In this setup, Case (a) and Case (b) examine the dual-branch architecture by removing the temporal branch and the spectral branch, respectively. Case (c) omits differential current extraction and instead uses raw current as the input, and a synthetic SAF set is generated directly from the raw signals, which is then used to fine-tune the pre-trained model. Case (d) establishes a baseline where ArcFusionNet utilizes raw current without being fine-tuned. Case (e) evaluates the model without generating synthetic SAF data for fine-tuning. Finally, Case (f) tests an alternative strategy where the pre-trained model is fine-tuned using a simple mixture of target-domain normal samples and source-domain SAF samples. The testing results are presented in Table 5, where the highlighted changes represent the performance deviations from the full proposed framework.

The experimental results indicate that both Case (a) and Case (b) suffer consistent performance degradations across all metrics. These systematic declines confirm the necessity of fusing temporal and spectral features for robust SAF detection. Moreover, discarding the temporal branch leads to a steeper deterioration than omitting the spectral branch. This highlights the dominant role of temporal features in the dual-branch model. Specifically, the temporal branch captures the complex morphological variations and local distortions of the differential currents, while spectral input provides supplementary harmonic information. Therefore, integrating temporal and spectral inputs is essential to establish a comprehensive feature space capable of distinguishing arcs from normal samples.

Furthermore, Case (c) demonstrates a significant reduction in the *FPR* to 0.3%. This decrease occurs because the fine-tuning process enables the model to learn the patterns of target-domain normal samples. However, the *FNR* rises to an unacceptable 25.2%, which is caused by two major issues stemming from the masking loads. First, the model struggles to extract highly discriminative features directly from the raw currents. Second, the overwhelming masking loads prevent the dCAM from isolating precise SAF-related sequences. Therefore, the synthetic SAF data generated under these conditions exhibits a distribution gap compared to real SAFs. Such degraded performance not only confirms the necessity of differential current extraction but also indicates that the dCAM requires integration with differential currents to achieve effective data synthesis.

Case (d) evaluates the baseline and reveals a severe performance degradation, where accuracy drops to 66.2% and the *FPR* rises to 83.2%. Fig. 12(a) presents a t-SNE visualization of the feature vectors extracted from the average pooling layer of ArcFusionNet pre-trained in the known scenario. It shows a severe domain shift where target-domain normal samples heavily overlap with the arc fault clusters, confirming that the model erroneously classifies normal samples as SAFs. Such a distribution stems from two issues. First, the low SNR of raw current prevents the model from extracting generalizable SAF features. Second, diverse masking loads introduce severe inherent distribution gaps across different domains. These combined effects cause the model to overfit to the scenario-specific background component in the source domain. Consequently, the model fails to accommodate the distributional shift in unknown scenarios. In contrast, Case (e) demonstrates the standalone effectiveness of differential current extraction by achieving a 90.8% accuracy without any fine-tuning. This process actively filters out dominant masking loads to improve the SNR. Fig. 12(b) visually verifies

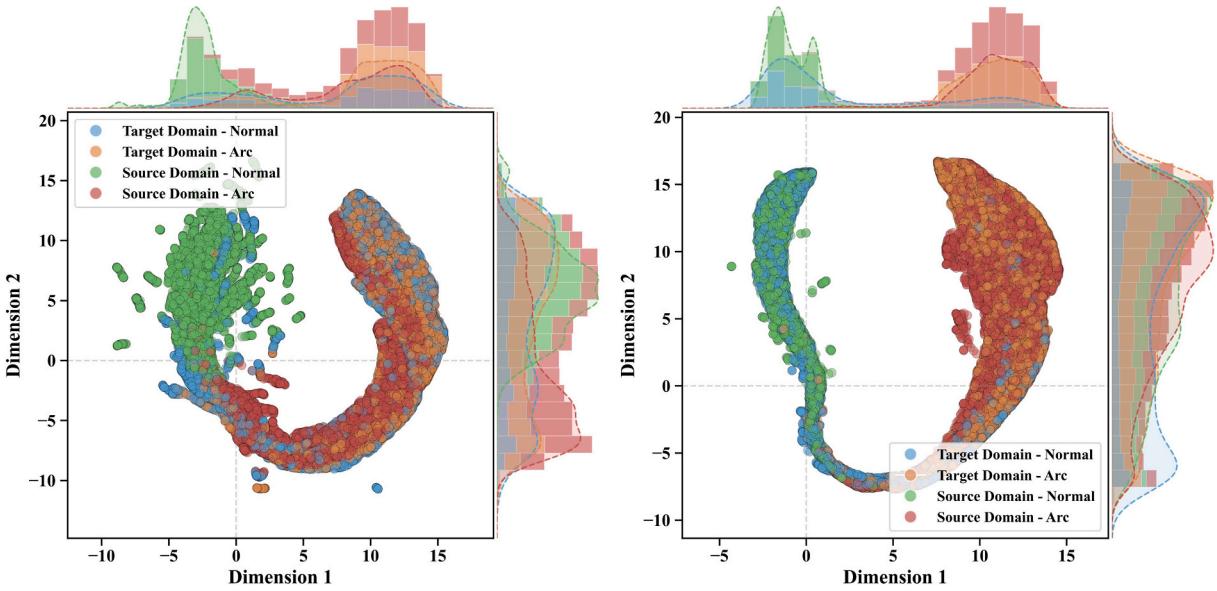


Fig. 12. t-SNE visualization of feature vectors extracted from the average pooling layer of arcfusionnet pre-trained in the source domain. (a) Derived from raw current. (b) Derived from differential current.

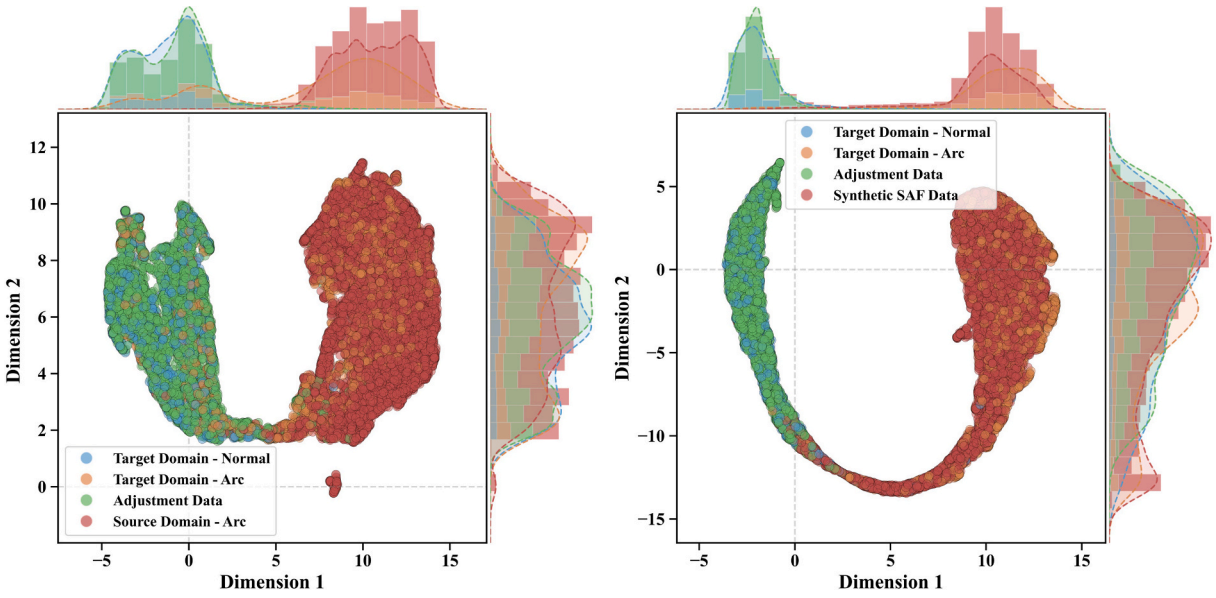


Fig. 13. t-SNE visualization of feature vectors extracted from the average pooling layer of fine-tuned ArcFusionNet based on differential current. (a) Mixed-domain fine-tuning. (b) Proposed fine-tuning.

this improvement through a highly structured t-SNE feature projection, which reveals smoothly aligned cross-domain distributions and a separation between normal and SAF samples. This confirms that differential currents facilitate the learning of robust domain-invariant features. Even so, a slight performance drop remains compared to the full proposed framework. The difference justifies the need for a fine-tuning step to handle target-domain residual noise.

Finally, Case (f) reveals that employing a simple mixture of target-domain normal data and source-domain arc data for fine-tuning yields an accuracy of 83.7% with an elevated *FNR* of 24.8%. This severe underperformance arises because the discrepancy between the arc and normal samples includes both the intrinsic arc signatures $F^{\text{arc_d}}$ and the residual noise components β^{d}. Since the target-domain noise β_B^{d} appears exclusively in the normal samples during the fine-tuning process, the model tends to associate these specific noise patterns with the normal class and may adopt a shortcut-learning strategy instead of extracting intrinsic SAF features. Therefore, when the model encounters real target-domain SAF samples containing β_B^{d} , it is easily misled into classifying them as normal samples. As shown in the t-SNE visualization in Fig. 13(a), when employing a mixed-domain fine-tuning strategy, the feature representations of the target-domain SAF samples are severely scattered across both the normal and arc feature spaces. To overcome this bottleneck, the proposed data synthesis method constructs a distribution approximating $\mathscr{D}\left(F_A^{\text{arc_d}}, \beta_B^{\text{d}}\right)$, ensuring that both training classes share the same target-domain noise β_B^{d}. It forces the model to focus exclusively on the intrinsic SAF features for reliable detection and actively prevents overfitting to background noise. Fig. 13(b) validates this by showing an alignment between the synthetic SAF data and the real target-domain SAF samples. The resulting feature space establishes a distinct separation boundary between most normal and SAF samples, demonstrating the effective extraction of robust domain-invariant

Table 6
Performance of benchmarks integrated with proposed components.

Ref.	Known scenario (validation set in source domain)						Unknown scenarios (testing set in target domain)					
	ACC (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)	ACC (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)
[11]+DC	88.5	9.5	14.9	87.7	87.8	87.8	77.7	12.3	39.6	76.6	74.1	75.3
[11]+DC+FT							83.8	16.8	15.3	82.4	84.0	83.2
[18]+DC	92.2	6.7	9.8	91.6	91.8	91.7	81.7	8.4	35.4	81.7	78.1	79.9
[18]+DC+FT							85.4	17.2	10.2	84.2	86.3	85.2
[22]+DC	97.9	1.9	2.4	97.7	97.8	97.8	88.3	5.4	22.6	88.6	86.0	87.3
[22]+DC+FT							95.6	4.8	3.6	95.0	95.8	95.4
[23]+DC	96.8	3.6	2.5	96.4	96.9	96.6	88.1	1.7	29.6	90.6	84.3	86.2
[23]+DC+FT							94.5	6.7	3.3	93.6	95.0	94.2
[24]+DC	94.1	4.6	8.0	93.8	93.7	93.7	84.5	10.0	24.9	83.7	82.5	83.1
[24]+DC+FT							95.4	5.1	3.8	94.7	95.6	95.1
[25]+DC	97.6	2.3	2.5	97.4	97.6	97.5	89.1	5.6	20.0	89.1	87.2	88.0
[25]+DC+FT							95.5	5.1	3.3	94.8	95.8	95.2
[26]+DC	97.1	2.7	3.2	96.8	97.0	96.9	87.2	8.6	20.2	86.5	85.6	86.1
[26]+DC+FT							94.2	6.3	5.1	93.4	94.3	93.8
[27]+DC	96.1	3.7	4.3	95.7	96.0	95.9	87.7	5.6	23.9	88.0	85.3	86.6
[27]+DC+FT							92.6	8.7	5.3	91.5	93.0	92.3
[28]+DC	95.6	4.0	5.1	95.2	95.5	95.3	83.3	11.1	26.4	82.3	81.3	81.8
[28]+DC+FT							88.3	13.8	8.0	87.1	89.1	88.1
Avg.Δ (+DC)							↑20.4	↑2.6	↓54.9	↑17.5	↑26.1	↑22.2
Avg.Δ (+DC+FT)	↓2.3	↑1.7	↑3.2	↓2.4	↓2.5	↓2.4	↑26.8	↑4.4	↓75.5	↑23.0	↑35.5	↑29.8

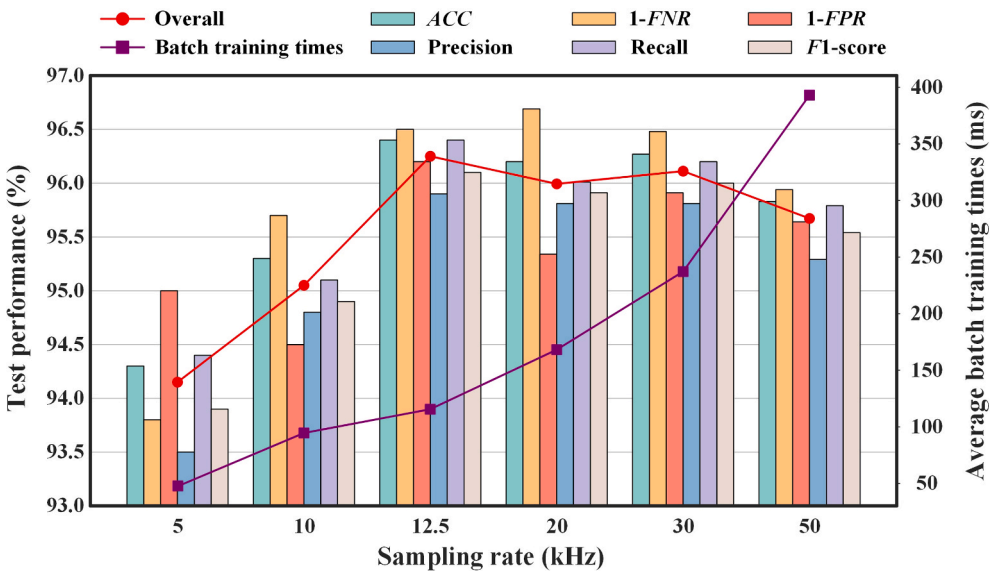


Fig. 14. Impact of sampling rate on detection performance and average batch training time.

representations.

4.4. Effectiveness of proposed components on benchmarks

To validate the universal applicability of the proposed core components, their effectiveness is systematically evaluated by integrating both the differential current extraction and the synthetic SAF generation methods into the nine benchmarks. This evaluation is conducted under two distinct learning configurations. In the first configuration (+DC), the models are trained using the extracted differential currents. In the second configuration (+DC + FT), the training process is further extended by fine-tuning the models on the synthetic SAF set together with the adjustment set. The resulting performance enhancements are detailed in Table 6.

The results confirm that the proposed components are not exclusive to ArcFusionNet but instead serve as universal enhancements for existing methods. Integrating the differential current extraction method alone yields substantial improvements across all benchmarks in the unknown scenario, increasing average accuracy by 20.4% and *F*1-score by 22.2%. These gains mainly result from a remarkable 54.9% reduction in the *FPR*. Notably, employing differential current causes a slight performance decrease in the known scenario. This occurs because removing dominant masking load currents forces the models to focus on generalizable arc features, thereby preventing overfitting to specific load patterns and eliminating deceptive over-optimism during known scenario evaluations.

Furthermore, the addition of the fine-tuning process provides further gains, increasing average accuracy by 26.8% and *F*1-score by 29.8%. Although the *FNR* rises slightly, this increase is outweighed by a consistent reduction in the *FPR*. Therefore, the overall detection performance improves across all metrics. It is also noteworthy that methods based on expert features exhibit less pronounced enhancements compared to end-to-end deep learning models, indicating that predefined features optimized for raw currents are insufficient to capture the signatures of differential currents.

4.5. Sensitivity analysis and discussion

To comprehensively validate the stability of the proposed framework, this section investigates the impact of different experimental settings. The analysis considers factors such as the data sampling rate, the number of input harmonic components, the source-domain data composition and so on.

4.5.1. Sensitivity analysis on data sampling rate

The sampling rate is a critical parameter influencing both feature representation and computational cost. Since the model takes a fixed one-cycle signal segment as input, the sampling rate directly determines the input dimensionality. To evaluate the effect of signal resolution, a sensitivity analysis is conducted over sampling rates ranging from 5 kHz to 50 kHz. The 32 input harmonics are dynamically re-evaluated for each configuration to ensure a fair comparison. The detection performance and average batch training time are shown in Fig. 14.

Increasing the sampling rate initially leads to a substantial enhancement in detection performance. As the rate rises from 5 kHz to 12.5 kHz, the overall accuracy improves from 94.3% to a peak of 96.4%, indicating that a sufficient sampling resolution facilitates the preservation of morphological details, which are essential for distinguishing arc fault features from background noise. The resulting performance gain justifies the proportional increase in computational cost. However, raising the sampling rate beyond 12.5 kHz results in a slight decline in overall accuracy rather than further improvements. This performance degradation occurs because excessively high sampling rates inevitably capture irrelevant high-frequency interference and environmental noise. Meanwhile, the increasing input dimensionality steadily raises the computational cost. Therefore, 12.5 kHz is identified as the optimal configuration, achieving a favorable balance between detection performance and computational cost. Notably, the detection accuracy remains at 94.3% even at the minimum rate of 5 kHz. This performance ensures reliable operation in scenarios with severely constrained resources, thereby demonstrating the deployment potential of the proposed framework.

4.5.2. Sensitivity analysis on source-domain data composition

Given the foundational role of source-domain data in model pretraining and SAF data synthesis, a sensitivity analysis is conducted to evaluate the impact of source-domain data composition on detection performance. Five distinct cases are designed to vary the quantity, complexity, and diversity of the source-domain training data. The first three cases investigate the effect of sample quantity. Cases (a), (b), and (c) restrict the sample quantity to 50%, 25%, and 10% of the original source-domain training set. Case (d) evaluates data complexity by filtering the source-domain training set to include only data completely free of masking loads. Case (e) constructs the training set by selectively

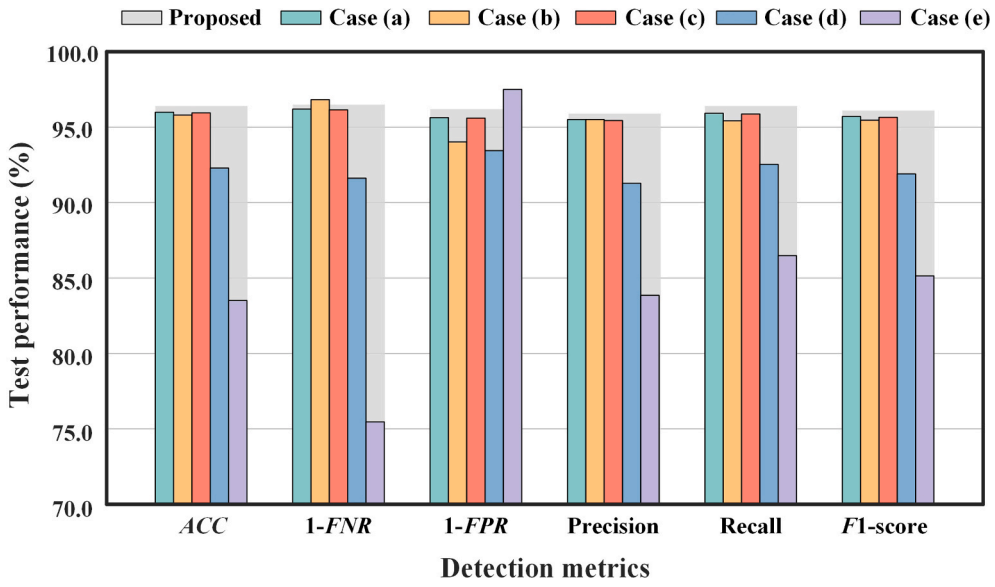


Fig. 15. The test performance with different source-domain data compositions.

retaining a single representative load from each load type. Specifically, the training set in Case (e) contains only the kettle, vacuum cleaner, fan, hairdryer, dimmer lamps, induction cooker, electric caldron, and microwave oven listed in Table 1. The resulting metrics under these cases are presented in Fig. 15.

The framework demonstrates exceptional robustness against variations in source data quantity. Even under the extreme constraint of Case (c), the model achieves a 96.0% accuracy and an $F1$-score of 95.7%. This stability indicates that even a severely restricted subset retains sufficient information to allow the model to extract comprehensive and representative intrinsic arc signatures $F_A^{\text{arc_d}}$ from the source domain. Furthermore, the low cost of the synthesis process allows the total amount of synthetic SAF data to remain consistent with the original experimental setup, thereby preserving the diversity of the synthetic samples. Although these results indicate that the method remains highly effective under severely limited source data availability, the original experiment still employs the full dataset to establish the highest possible upper bound for detection reliability.

In the data complexity evaluation of Case (d), a noticeable performance decline is observed relative to the original framework. The complete absence of background noise removes a crucial regularization effect during pre-training, which may cause the model to overfit to irrelevant SAF features and subsequently mislead the dCAM module into extracting inaccurate SAF-related sequences. Remarkably, even when the SAF-related sequence extraction process becomes suboptimal, the majority of correctly extracted sequences still guarantee the high quality of the generated SAF data, allowing the model to achieve a competitive accuracy of 92.3%.

Under the extreme diversity constraint of Case (e), the model exhibits a performance degradation, with the accuracy decreasing to 83.5%. Restricting the training data to a single representative load for each load type hinders the extraction of comprehensive arc signatures $F_A^{\text{arc_d}}$, causing the model to miss the rich signal variations inherent in different loads. Despite these severe limitations, the proposed framework still avoids a catastrophic performance collapse. This residual effectiveness can be attributed to the shared operating mechanisms among loads of the same type, which preserve a certain degree of similarity in their intrinsic SAF signatures. Overall, maintaining rich load diversity in the source domain is essential for the proposed framework to ensure robust detection performance in unknown scenarios.

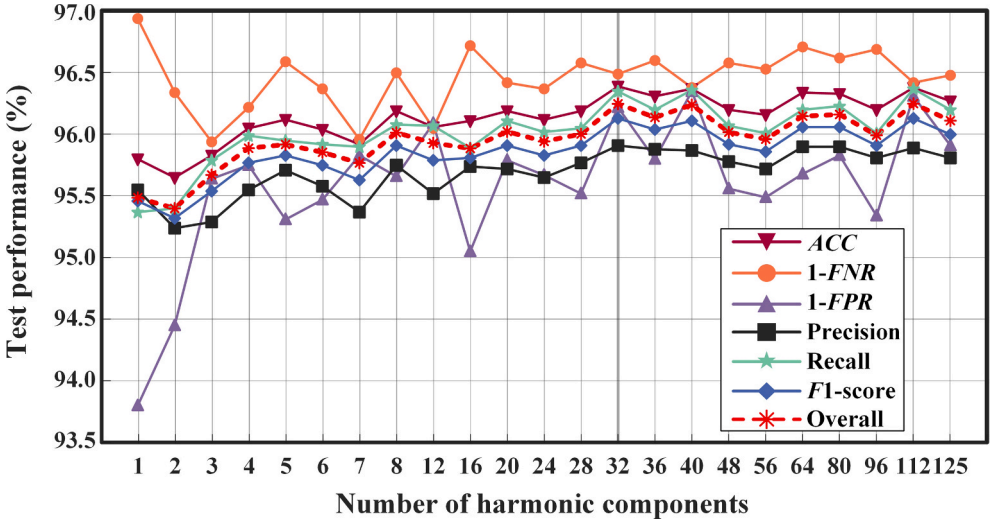


Fig. 16. The test performance with different numbers of harmonic components.

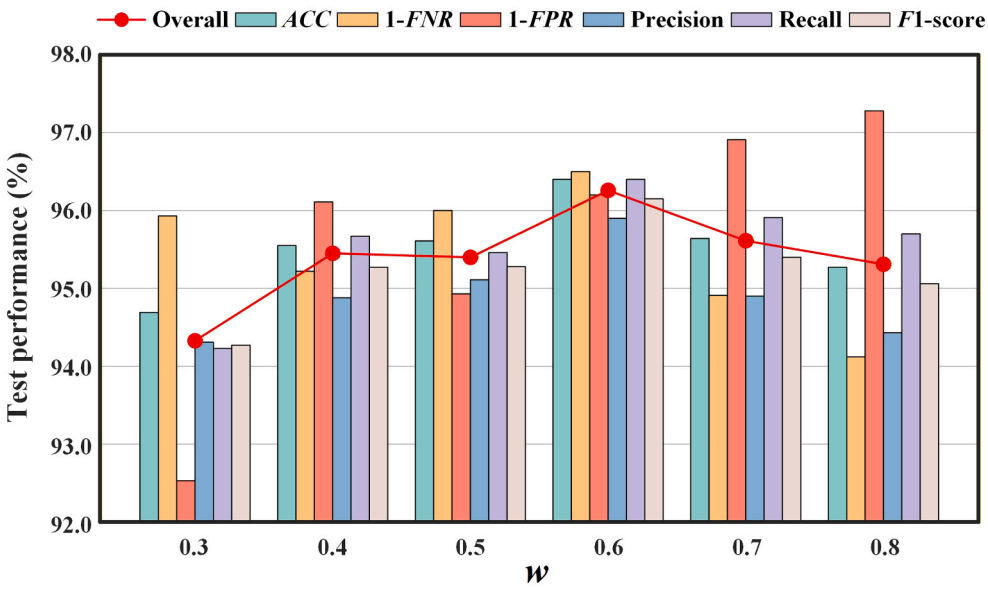


Fig. 17. Impact of the weighting factor on detection performance in the unknown scenario.

4.5.3. Sensitivity analysis on the number of input harmonic components

To evaluate the impact of input harmonic dimensionality on detection performance, ArcFusionNet is pre-trained and fine-tuned using different numbers of top-ranked harmonic components as the input to the spectral branch. These components are cumulatively selected according to their importance scores shown in Fig. 11, and the testing results are presented in Fig. 16.

As the number of selected high-importance components increases from 1 to 32, the overall detection metrics steadily improve toward their peak values. This continuous improvement validates the feature ranking mechanism, confirming that accumulating these top-ranked harmonics constructs the most comprehensive and informative SAF signatures. However, further including lower-ranked harmonics beyond this optimal threshold leads to slight performance degradation. Since these additional components correspond to the lowest importance scores, they provide minimal discriminative information. Furthermore, incorporating these components introduces redundant load-specific noise, which compromises the generalization capability of the model. Therefore, retaining the top 32 harmonic components is identified as the optimal configuration in the experiments. This dimensionality captures the maximum discriminative information while actively filtering out disruptive redundant noise and avoiding unnecessary computational burden, thereby ensuring a robust and efficient feature representation.

4.5.4. Sensitivity analysis on hyperparameter w

The hyperparameter w in Eq. (9) governs the trade-off between the injected SAF signatures $F_A^{\mathrm{arc_d}}$ and the target-domain background noise β_B^{d} during SAF data synthesis. To determine the optimal balance between these two components, a sensitivity analysis is conducted by varying w from 0.3 to 0.8. The corresponding impact on the detection performance in the unknown scenario is illustrated in Fig. 17.

Initially, setting w to a low value of 0.3 causes the synthetic data to be primarily dominated by the target-domain background noise. In this case, the injected fault signatures become too subtle to be reliably extracted, yielding an accuracy of 94.7% and a relatively high *FPR* of 7.5%. As w gradually increases, the distinctive arc signatures become more pronounced, leading to a steady improvement in most evaluation metrics. The overall performance reaches its peak at $w = 0.6$, and the model achieves the highest accuracy and *F*1-score while simultaneously reducing the *FNR* to 3.5%. This optimal balance ensures that the model captures the essential fault features without being overwhelmed by residual noise interference. However, raising w to 0.8 degrades the overall detection capability. Although the *FPR* reaches a minimum of 2.7%, this improvement comes at the cost of a substantial increase in the *FNR*, indicating excessively dominant injected SAF signatures distort the realistic noise distribution of the target domain. The model consequently learns exaggerated features and fails to recognize weaker and real arcing events. Hence, $w = 0.6$ is selected as the optimal synthesis parameter in the experiments, as it prevents both the under-learning of subtle faults and the overfitting to artificial signature distortions.

Table 7
Detection performance under various SNR conditions.

SNR (dB)	*ACC* (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)
30	94.5	5.9	4.8	94.6	94.5	94.6
25	94.4	6.9	3.4	94.7	94.4	94.5
20	93.6	7.6	4.3	93.9	93.6	93.8
15	92.1	9.8	4.6	92.6	92.1	92.3
10	90.4	10.2	8.5	90.8	90.4	90.6
5	88.3	10.1	14.4	87.3	87.8	87.5

4.6. Analysis of robustness against background noise

The practical applicability of the framework also relies on its robustness to the residual background noise. After dominant masking-load interference is suppressed by differential current extraction, the remaining masking-load effects are mainly reflected as residual noise in the extracted differential currents. To assess the proposed framework's robustness across various SNR conditions, additive white Gaussian noise is superimposed onto the testing set in the unknown scenario, providing an indirect assessment of the framework's resilience against the noise-like component of masking-load interference. The SNR levels range from 30 dB down to an extremely severe level of 5 dB, and the corresponding detection performance is presented in Table 7.

The results demonstrate that the proposed framework maintains high stability under relatively high SNR conditions. Specifically, at 30 dB and 25 dB, the detection accuracy exceeds 94.4%, indicating that the typical intensity of residual noise imposes a negligible impact on the model's detection capability. As the noise interference intensifies, the performance declines gradually while remaining at a reliable level. Even when the SNR drops to 10 dB, the model achieves an accuracy of 90.4%,

Table 8
Performance Comparison of Different Models.

Model	*ACC* (%)	*FNR* (%)	*FPR* (%)	Precision (%)	Recall (%)	*F*1-score (%)
Teacher	96.4	3.5	3.8	95.9	96.4	96.1
Student (KDKR)	95.7	5.0	3.1	95.0	96.0	95.4
Student (baseline)	91.5	8.7	8.3	90.9	91.5	91.2

demonstrating strong resilience to severe background noise. Under the extreme condition of 5 dB, the model still achieves an acceptable accuracy of 88.3%. However, the *FPR* surges to 14.4%, as high-intensity noise tends to cause the model to erroneously classify noisy normal signals as faults. Despite this limitation under extreme conditions, the overall results confirm that the framework is sufficiently robust to operate reliably in most practical environments, where residual noise typically remains below such severe levels. This also provides indirect evidence for the reliability against the unpredictable residual noise remaining after the differential current extraction of more diverse and complex masking-load conditions.

4.7. Lightweight network evaluation

According to the IEC 62606 standard, a reliable SAF detector must respond within a specified time range of 0.12 s–1 s [34]. To verify the effectiveness of the KDKR strategy in reducing the computational cost to meet this timing constraint, a comparative experiment is conducted. Specifically, the full-sized ArcFusionNet serves as the teacher network, and the student network inherits the teacher's architecture, with the channel dimensions of the convolutional layers in the temporal and spectral branches compressed to 1/8 and 1/4 of their respective sizes in the teacher network. To highlight the performance gain brought by knowledge distillation, the KDKR-trained student is evaluated against a baseline model, which shares the same lightweight architecture but is trained from scratch. The detection performance comparison is presented in Table 8. The results reveal that the student network distilled via KDKR achieves a robust accuracy of 95.7%. In contrast, the baseline student trained from scratch exhibits a performance degradation due to its inherently limited representation capacity, with the accuracy dropping to 91.5%. This discrepancy indicates that KDKR effectively mitigates the performance loss caused by reduced model complexity, enabling the lightweight network to achieve performance highly comparable to that of its teacher.

To evaluate its deployment efficiency on edge devices, the student network is quantized through the Open Neural Network Exchange format, followed by a comprehensive assessment of deployment metrics, including the number of model parameters, model size, floating point operations (FLOPs), multiply-accumulate operations (MACs), memory access cost (MAC) and detection runtime. Specifically, the detection runtime is measured on the Raspberry Pi 4B by averaging the results over 1000 independent test runs. The detailed comparisons between the full-sized teacher and the lightweight student networks are summarized in Table 9. The quantitative results show that the proposed KDKR incurs only a marginal loss in detection performance while substantially reducing deployment requirements. Specifically, it reduces the number of model parameters by 94.1%, model size by 91.6%, both FLOPs and MACs by 95.1%, while reducing the memory access cost by 20.8%. Moreover, the detection time on the Raspberry Pi 4B is reduced from 303.6 ms to 59.8 ms, meeting the time requirements for SAF detection set by the IEC 62606 standard. This trade-off between accuracy and deployment efficiency demonstrates the effectiveness of the proposed KDKR strategy, and enables the proposed method to be deployed as a practical SAF detection solution on resource-constrained edge devices.

5. Conclusions

Masking loads severely interfere with the extraction of domain-invariant SAF features, and their unpredictable combinations result in substantial data distribution shifts between different domains. Conventional methods struggle to achieve robust non-intrusive SAF detection in unknown scenarios, where real target-domain SAF data are unavailable for model adaptation. To solve this problem, this paper proposes a holistic non-intrusive detection framework driven by two synergistic innovations. First, a differential current extraction method is proposed to obtain signals that preserve intrinsic fault information while reducing the influence of masking loads, inherently facilitating the extraction of robust fault features and reducing the distribution gap across different scenarios. Second, to address the SAF data scarcity in the target domain, an interpretable dCAM-based SAF synthesis method is developed to generate realistic fault samples, further reducing the remaining domain discrepancy without requiring actual fault data. With the proposed temporal-spectral fusion network, ArcFusionNet, serving as the effective classifier, this framework achieves an accuracy of 96.4% in the unknown scenario, representing a significant improvement from the 66.2% accuracy of the baseline model. Furthermore, comprehensive experiments validate the framework's broad universality across existing models, as well as its strong robustness and stability under severe background noise and restricted source-domain data. For practical edge deployment, ArcFusionNet is distilled into a lightweight model using KDKR, preserving a 95.7% accuracy while reducing model size and FLOPs by over 90%. The lightweight network achieves a detection time of 59.8 ms on the Raspberry Pi 4B, highlighting its potential for wide-scale practical deployment.

While the proposed method achieves robust detection performance, practical limitations remain since it currently targets steady-state currents under simplified load switching conditions. Therefore, its performance in processing transient currents and continuously variable loads remains to be validated. Future research will focus on developing more robust LSCP and SSAP detectors for complex switching scenarios and extending the framework to handle transient and continuously variable currents. Its applicability in diverse and complex real-world residential power systems will also be further evaluated under a wider variety of masking loads.

CRediT authorship contribution statement

Wenpeng Luan: Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation. **Hanju Cai:** Writing – review & editing, Writing – original draft,

Table 9
Metrics of Teacher and Student Networks.

Model	Params (K)	Size (KB)	FLOPs (M)	MACs (M)	MAC (MB)	Time (ms)
Teacher	904.5	3610.0	1130.2	564.5	12.0	303.6
Student	**53.0** (↓94.1%)	**302.0** (↓91.6%)	**55.9** (↓95.1%)	**27.7** (↓95.1%)	**9.5** (↓20.8%)	**59.8** (↓80.3%)

Visualization, Validation, Software, Methodology, Investigation, Data curation. **Bochao Zhao:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis. **Yang Han:** Writing – review & editing, Validation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Y. Zhang, H.C. Chen, Z. Li, C. Jia, Y. Du, K. Zhang, H. Wei, Lightweight ac arc fault detection method by integration of event-based load classification, IEEE Trans. Ind. Electron. 71 (4) (2024) 4130–4140.
[2] H. Gao, Z. Wang, C. Han, A. Tang, F. Guo, B. Li, Feature extraction method of series arc fault occurred in three-phase motor with inverter circuit, IEEE Trans. Power Electron. 37 (9) (2022) 11164–11173.
[3] S. Hwang, B. Kim, M. Kim, H.-P. Park, Ac series arc fault detection for wind power systems based on phase lock loop with time and frequency domain analyses, IEEE Trans. Power Electron. 39 (10) (2024) 12446–12455.
[4] Q. Gong, Q. Gao, K. Peng, Y. Liu, Y. Jiang, Low-voltage AC series arc fault identification method based on the randomness of differential voltage at double-ended monitoring points, IEEE Trans. Instrum. Meas. 74 (2025) 1–17.
[5] S. Jia, Y. Jia, X. Luo, S. Ji, L. Lv, Optical signal characteristics of power frequency arc in transformer oil, IEEE Trans. Power Del. 39 (1) (2024) 675–678.
[6] Y. Zhang, K. Jia, Y. Li, T. Bi, Incipient fault detection-based active protection of photovoltaic dc integration system, IEEE Trans. Power Del. 39 (2) (2024) 1198–1208.
[7] K. Li, S. Zhao, Y. Wang, A planar location method for dc arc faults using dual radiation detection points and dann, IEEE Trans. Instrum. Meas. 69 (8) (2020) 5478–5487.
[8] J.C. Kim, D.O. Neacsu, R. Ball, B. Lehman, Clearing series ac arc faults and avoiding false alarms using only voltage waveforms, IEEE Trans. Power Del. 35 (2) (2020) 946–956.
[9] Z. He, Z. Xu, H. Zhao, W. Li, Y. Zhen, W. Ning, Detecting series arc faults using high-frequency components of branch voltage coupling signal, IEEE Trans. Instrum. Meas. 73 (2024) 1–13.
[10] G. Zhou, L. Huang, Z. Li, H. Tian, B. Zhang, M. Fu, Y. Feng, C. Huang, Intever public database for arcing event detection: feature analysis, benchmark test, and multi-scale cnn application, IEEE Trans. Instrum. Meas. 70 (2021) 1–15.
[11] W. Dai, X. Zhou, Z. Sun, Q. Miao, G. Zhai, Series ac arc fault detection method based on l2/l1 norm and classification algorithm, IEEE Sens. J. 24 (10) (2024) 16661–16672.
[12] J. Lezama, P. Schweitzer, E. Tisserand, J.-B. Humbert, S. Weber, P. Joyeux, An embedded system for ac series arc detection by inter-period correlations of current, Electr. Power Syst. Res. 129 (2015) 227–234.
[13] W. Luan, J. Lin, B. Liu, B. Zhao, Arc fault detection and identification via non-intrusive current disaggregation, Electr. Power Syst. Res. 210 (2022) 108113.
[14] H. Tong, X. Zeng, K. Yu, J. Mu, C. Luo, B. Liu, Research on the identification method of series arc fault based on the feature sensitivity analysis and the TVA coefficient optimized random forest, IEEE Trans. Power Del. 39 (2) (2024) 751–762.
[15] Y. Wang, F. Zhang, X. Zhang, S. Zhang, Series AC arc fault detection method based on hybrid time and frequency analysis and fully connected neural network, IEEE Trans. Ind. Informat. 15 (12) (2019) 6210–6219.
[16] G. Artale, A. Cataliotti, V. Cosentino, D. Di Cara, S. Nuccio, G. Tine, Arc fault detection method based on CZT low-frequency harmonic current analysis, IEEE Trans. Instrum. Meas. 66 (5) (2017) 888–896.
[17] P. Qi, S. Jovanovic, J. Lezama, P. Schweitzer, Discrete wavelet transform optimal parameters estimation for arc fault detection in low-voltage residential power networks, Electr. Power Syst. Res. 143 (2017) 130–139.
[18] B. Li, J. Wu, Low-voltage arc fault identification using a hybrid method based on improved Salp Swarm algorithm-variational mode decomposition-random forest, IEEE Access 12 (2024) 15410–15418.
[19] L. Du, Y. Shen, Z. Xu, D. Chen, Enhancing arc fault diagnosis method using feature selection strategy based on feature clustering and maximal information coefficient, IEEE Trans. Ind. Appl. 60 (2) (2024) 3006–3017.
[20] L. Du, Z. Xu, H. Chen, D. Chen, Feature selection-based low-voltage AC arc fault diagnosis method, IEEE Trans. Instrum. Meas. 72 (2023) 1–12.
[21] J. Jiang, W. Li, Z. Wen, Y. Bie, H. Schwarz, C. Zhang, Series arc fault detection based on random forest and deep neural network, IEEE Sens. J. 21 (15) (Aug. 2021) 17171–17179.
[22] Y. Wang, L. Hou, K.C. Paul, Y. Ban, C. Chen, T. Zhao, Arcnet: series AC arc fault detection based on raw current and convolutional neural network, IEEE Trans. Ind. Informat. 18 (1) (2022) 77–86.
[23] A. Tang, Z. Wang, S. Tian, H. Gao, Y. Gao, F. Guo, Series arc fault identification method based on lightweight convolutional neural network, IEEE Access 12 (2024) 5851–5863.
[24] W. Dai, X. Zhou, Z. Sun, G. Zhai, Series alternating current arc fault detection method based on relative position matrix and deep convolutional neural network, Eng. Appl. Artif. Intell. 136 (2024).
[25] S. Zhang, N. Qu, T. Zheng, C. Hu, Series arc fault detection based on wavelet compression reconstruction data enhancement and deep residual network, IEEE Trans. Instrum. Meas. 71 (2022) 1–9.
[26] T. Zhang, R. Zhang, H. Wang, R. Tu, K. Yang, Series AC arc fault diagnosis based on data enhancement and adaptive asymmetric convolutional neural network, IEEE Sens. J. 21 (18) (2021) 20665–20673.
[27] H. Liu, J. Li, W. Wang, S. Lu, Arc-1DCNN: an enhanced model for series arc fault detection, Measurement 234 (2024) 114814.
[28] X. Zhou, W. Dai, Z. Sun, G. Zhai, Difference matrix and self-attention mechanism for series AC arc fault detection, IEEE Trans. Instrum. Meas. 74 (2025) 1–12.
[29] R. Jiang, G. Bao, Q. Hong, C.D. Booth, Machine learning approach to detect arc faults based on regular coupling features, IEEE Trans. Ind. Informat. 19 (3) (2023) 2761–2771.
[30] C. Jia, Z. Li, Y. Zhang, J. Cao, H. Chen, X. Zhao, Y. Du, Q.S. Cheng, N. Ma, F. Qiu, Serial arc-fault restriction survey with standard-listed products in the chinese market, IEEE Trans. Ind. Appl. 59 (6) (2023) 7453–7461.
[31] M. Tan, Y. Gong, L. Wang, K. Li, J. Tong, Y. Su, LFArc-PFE: a series arc fault detection method based on low-frequency current data and perturbation feature extraction, IEEE Trans. Instrum. Meas. 74 (2025) 1–11.
[32] UL 1699: UL Standard for safety Arc-Fault Circuit-Interrupters, 2019.
[33] M. Zhao, S. Zhong, X. Fu, B. Tang, M. Pecht, Deep residual shrinkage networks for fault diagnosis, IEEE Trans. Ind. Informat. 16 (7) (July 2020) 4681–4690.
[34] General Requirements for Arc Fault Detection Devices, IEC 62606, 2017.
[35] P. Boniol, M. Meftah, E. Remy, T. Palpanas, Dcam: dimension-wise class activation map for explaining multivariate data series classification, in: Proc. of the ACM SIGMOD Int. Conf. on Management of Data, 2022, pp. 1175–1189.
[36] P. Chen, S. Liu, H. Zhao, J. Jia, Distilling knowledge via knowledge review, in: Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, 2021, pp. 5006–5015.